



A double perturbation method of postbuckling analysis in 2D curved beams for assembly of 3D ribbon-shaped structures



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ABSTRACT

Mechanically-guided 3D assembly based on controlled, compressive buckling represents a promising, emerging approach for forming complex 3D mesostructures in advanced materials. Due to the versatile applicability to a broad set of material types (including device-grade single-crystal silicon) over length scales from nanometers to centimeters, a wide range of novel applications have been demonstrated in soft electronic systems, interactive bio-interfaces as well as tunable electromagnetic devices. Previously reported 3D designs relied mainly on finite element analyses (FEA) as a guide, but the massive numerical simulations and computational efforts necessary to obtain the assembly parameters for a targeted 3D geometry prevent rapid exploration of engineering options. A systematic understanding of the relationship between a 3D shape and the associated parameters for assembly requires the development of a general theory for the postbuckling process. In this paper, a double perturbation method is established for the postbuckling analyses of planar curved beams, of direct relevance to the assembly of ribbon-shaped 3D mesostructures. By introducing two perturbation parameters related to the initial configuration and the deformation, the highly nonlinear governing equations can be transformed into a series of solvable, linear equations that give analytic solutions to the displacements and curvatures during postbuckling. Systematic analyses of postbuckling in three representative ribbon shapes (sinusoidal, polynomial and arc configurations) illustrate the validity of the theoretical method, through comparisons to the results of experiment and FEA. These results shed light on the relationship between the important deformation quantities (e.g., mode ratio and maximum strain) and the assembly parameters (e.g., initial configuration and the applied strain). This double perturbation method provides an attractive route to the inverse design of ribbon-shaped 3D geometries, as demonstrated in a class of helical mesostructures.

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1. Introduction

Development of routes to 3D micro/nano-structures of advanced materials is of increasing interest, due to potential applications across a broad range of engineered systems, from biomedical devices (Karnaushenko et al., 2015; Kim et al., 2011; Leong et al., 2009; Tian et al., 2012), to microelectromechanical components (Bishop et al., 2012; Wood, 2014), photonic/optoelectronic platforms (Lee et al., 2014; Songmuang et al., 2007), and high performance electronics (Jang et al., 2017; Lewis and Ahn, 2015; Russo et al., 2011; Xu et al., 2011; Zhou et al., 2017). Several types of assembly/fabrication techniques can be considered, including those that rely on volumetric optical exposures (Arpin et al., 2010; Jang et al., 2007; Kong et al., 2014; Schaedler et al., 2011), nozzle-based filamentary extrusions (Cesarano et al., 1998; Lewis and Gratson, 2004; Murphy and Atala, 2014; Song et al., 1999; Truby and Lewis, 2016), fluidic self-assembly (Crane et al., 2013; Guo et al., 2009), and stress-induced bending/twisting (Bishop et al., 2012; Filipov et al., 2016, 2015; Lestringant and Audoly, 2017; Liu et al., 2014; Ni and Soh, 2014; Pan et al., 2014; Shenoy and Gracias, 2012). Although these techniques offer powerful capabilities in forming 3D mesostructures, each has limitations in terms of the accessible range of material types, realizable geometries or feature sizes (Zhang et al., 2017).

Recently, a mechanically guided 3D assembly approach that exploits controlled, compressive buckling of patterned 2D ribbons/membrane was proposed and demonstrated by Xu et al. (2015) and Zhang et al. (2015). This approach is fully compatible with nearly all established planar microsystems technologies, including those associated with the semiconductor industry, and is applicable to a broad set of high-performance materials (e.g., single-crystal silicon, metal, and polymer) and even fully-integrated electronic/optoelectronic systems, over length scales from nanometers to centimeters. Fig. 1 provides an example that illustrates the assembly process, in which a patterned 2D structure, which we refer to as a precursor, bonds to a prestretched elastomer substrate at selective locations. Initial release of the prestretch triggers the lateral buckling of the 2D precursors, with coupled flexural-torsional deformation modes. Full release completes the assembly process by transforming the 2D precursors into 3D structures, through a combination of well-controlled spatial translational/rotational motions and bending/twisting deformations. Due to the high level of design flexibility, i.e., either through the geometry of the 2D precursor and its bonding sites or the assembly platform itself, a variety of complex 3D geometries can be realized (Nan et al., 2017; Xu et al., 2015; Xue et al., 2017; Yan et al., 2016a, b; Zhang et al., 2017), not only in advanced materials but also in fully integrated device systems for monitoring human health (Jang et al., 2017), interacting with the growth of biological tissues (McCracken et al., 2017) and performing wireless communications (Yan et al., 2016a). Previous studies of this assembly approach rely mainly on FEA numerical simulations as a basis to guide the design process. A significant drawback is in the considerable amount of computational resources required, due to a large number of design parameters and the convergence issues in stability analyses.

A systematic understanding of the relationship between the final 3D shape and fabrication parameters requires the development of a general theory for the postbuckling process. 3D assembly for representative classes of slender, ribbon-type geometries involves the postbuckling analysis of planar curved beams. Tremendous progress has been made on the theoretical development of buckling and initial postbuckling in planar beams (Reis, 2015), dating back to Euler's foundational work (Euler, 1744). Koiter developed a systematic method for the initial postbuckling analysis (Koiter, 1963; van der Heijden, 2009), which was further extended and explored by several researchers (Budiansky, 1974; Hui, 1988; Su et al., 2012). Dias and Audoly (2015) established a reduced 1D theory of elastic developable ribbons under large deformations, and used it in the analysis of the buckling of a cylindrical ribbon. Recently, Fan et al. (2017) utilized a perturbation method in the postbuckling analysis of planar curved beams, and obtained analytical solutions to three special models, which cannot be generalized to other boundary conditions or geometries. In other work, many useful results on flexural-torsional buckling have been obtained by semi-analytical or numerical methods (Audoly and Seffen, 2015; Chen et al., 2013; Jawed et al., 2015; Liu and Lu, 2016; Liu et al., 2016; Trahair, 1993; Trahair and Papangelis, 1987; Xue et al., 2017). All such previous studies are limited to planar beams with constant initial curvature (i.e. straight beams or arches), due to the geometrical nonlinearity of the postbuckling analyses. These existing models, however, cannot be extended directly to the lateral postbuckling of planar beams with varying initial curvatures, as required in the context of the 3D assembly processes studied here.

In this paper, we establish a double perturbation method of postbuckling analyses in general planar curved beams, by introducing two small parameters for the deformation and the initial configuration respectively. This method significantly simplifies the nonlinear equilibrium differential equations of flexural-torsional postbuckling, and yields analytic solutions to the deformations and maximum strains. Experiments and FEA studies validate the predictions of this approach. Demonstration of an inverse design that identifies optimal 2D precursors to yield desired 3D helical mesostructures illustrates the utility of the analytic results. This paper is organized as follows: Section 2 introduces the formulation of the double perturbation method and its application to two representative loading conditions (i.e., the end-to-end compression and the oblique compression) in 3D assembly. Section 3 presents the results of postbuckling deformations for three representative geometries (i.e., sinusoidal-shaped, polynomial-shaped and arc-shaped beams), and investigates the truncation errors of the analytic results. This section also discusses the extension of the theoretical model to an inverse design problem for the widely used helical mesostructures with different shapes.

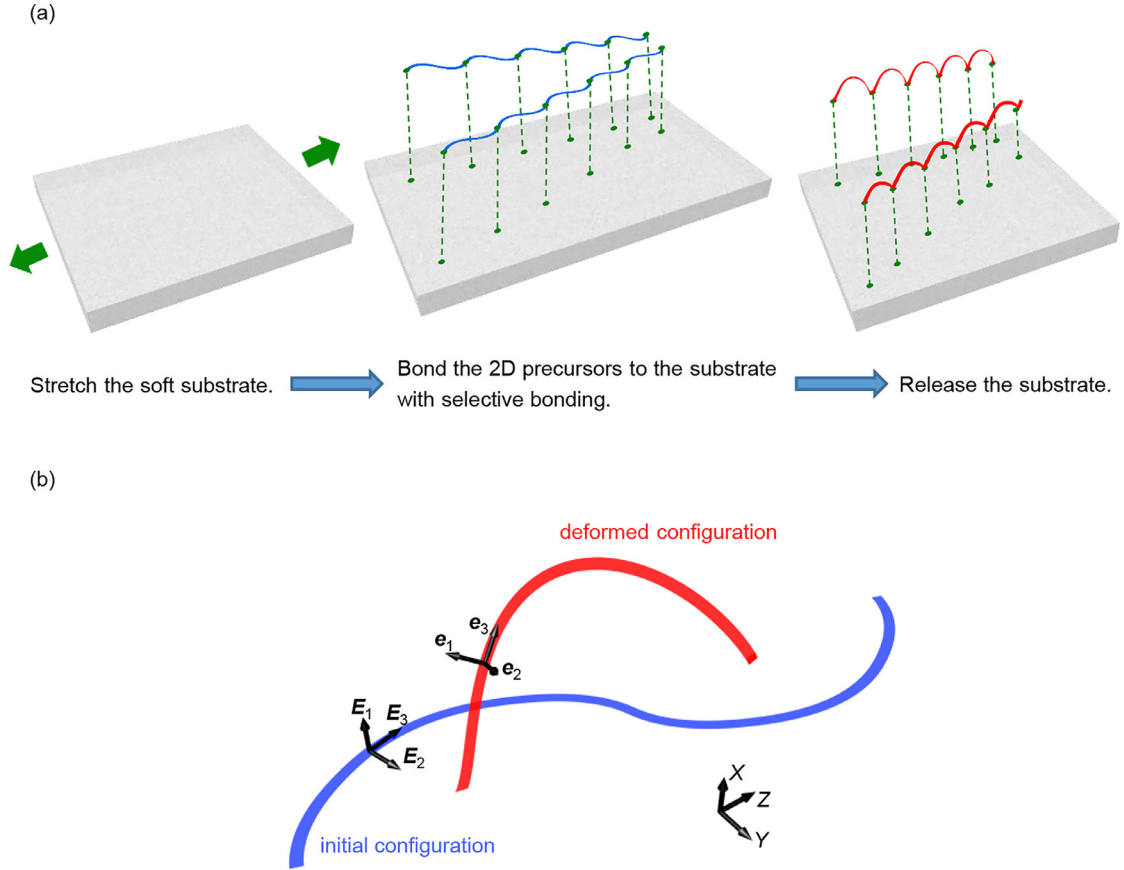


Fig. 1. Schematic illustrations of (a) the processes for the assembly of 3D microstructure by compressive buckling and (b) different coordinate systems associated with the buckling deformations of a planar curved beam.

2. Formulation of the method

As described in the introduction section, this paper focuses on buckling-guided assembly in a class of ribbon-type geometries that can be regarded as slender, curved beams, such that the Euler-Bernoulli assumption holds. Here, the deformed configuration of the curved beam can be fully characterized with the displacements of the centroid line and the twist angle of the cross section.

2.1. Basic equations

Consider a planar curved beam in the YZ plane, as illustrated in Fig. 1(b). The arc-length coordinate of the centroid line is denoted by S . Let the orthonormal material frame $\mathbf{E}_i$ ($i = 1, 2, 3$) be the local triad vectors at the centroid line, in which $\mathbf{E}_1$ and $\mathbf{E}_2$ are along the principal axes of the cross section, and $\mathbf{E}_3$ is tangential to the centroid line.

Under an external loading, the planar curved beam deforms into a spatial configuration with an arc-length coordinate s , and the elongation of the centroid line can be written as

$$\lambda = \frac{ds}{dS} = \left\{ 1 + 2 \left(\frac{dU_3}{dS} + U_2 K_1 \right) + \left[\left(\frac{dU_1}{dS} \right)^2 + \left(\frac{dU_2}{dS} - U_3 K_1 \right)^2 + \left(\frac{dU_3}{dS} + U_2 K_1 \right)^2 \right] \right\}^{\frac{1}{2}}, \quad (1)$$

in which $U_i = \mathbf{U} \cdot \mathbf{E}_i$ ($i = 1, 2, 3$) are the components of displacement vector of the centroid line $\mathbf{U}$, and K_1 is the initial curvature. During the deformation, the material frame $\mathbf{E}_i$ ($i = 1, 2, 3$) moves to the orthonormal frame $\mathbf{e}_i$ ($i = 1, 2, 3$) that is defined on the deformed configuration. Then, the curvature vector $\hat{\mathbf{k}}$ of deformed beam can be related to $\mathbf{e}_i$ by

$$\frac{d\mathbf{e}_i}{ds} = \hat{\mathbf{k}} \times \mathbf{e}_i (i = 1, 2, 3), \quad (2)$$

where $\hat{\mathbf{k}} = \hat{\kappa}_i \mathbf{e}_i$, with $\hat{\kappa}_1$ and $\hat{\kappa}_2$ denoting the bending curvatures, and $\hat{\kappa}_3$ denoting the twisting curvature along the centroid line. The twist angle ϕ relates to the twisting curvature $\hat{\kappa}_3$ by

$$\phi = \int \hat{\kappa}_3 dS. \quad (3)$$

It is noteworthy that the twist angle here is a quantity representing the spatial integral of torsional deformation ($\hat{\kappa}_3$) between two different cross sections, rather than an absolute angle of a certain cross section. In the finite deformation framework, the direction of the cross section in the deformed configuration, $\mathbf{e}_i$ ($i = 1, 2, 3$), is related to the material frame $\mathbf{E}_i$ ($i = 1, 2, 3$) by the cosine coefficients a_{ij} as

$$\mathbf{e}_i = a_{ij} \mathbf{E}_j \quad (i, j = 1, 2, 3). \quad (4)$$

According to Fan et al. (2017), the curvatures $\hat{\kappa}_i$ and cosine coefficients a_{ij} can be expressed in terms of generalized displacements (i.e., U_i and ϕ) as

$$\begin{Bmatrix} \hat{\kappa}_1 \\ \hat{\kappa}_2 \\ \hat{\kappa}_3 \end{Bmatrix} = \begin{Bmatrix} K_1 \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} -\frac{d^2 U_2}{ds^2} + \frac{d(U_3 K_1)}{ds} \\ \frac{d^2 U_1}{ds^2} - \psi K_1 \\ \frac{d\phi}{ds} \end{Bmatrix} + \{\hat{\kappa}_i^{(2)}\} + \{\dots\}, \quad (5)$$

$$\{a_{ij}\} = \begin{Bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{Bmatrix} + \begin{Bmatrix} 0 & \psi & -\frac{dU_1}{ds} \\ -\psi & 0 & -\frac{dU_2}{ds} + U_3 K_1 \\ \frac{dU_1}{ds} & \frac{dU_2}{ds} - U_3 K_1 & 0 \end{Bmatrix} + \{a_{ij}^{(2)}\} + \{\dots\} \quad (i, j = 1, 2, 3), \quad (6)$$

where $\psi = \int (\hat{\kappa}_3 - K_1 \frac{dU_1}{ds}) dS = \phi - \int K_1 \frac{dU_1}{ds} dS$, and the superscripts $\langle 2 \rangle$ in $\hat{\kappa}_i^{(2)}$ and $a_{ij}^{(2)}$ refer to the 2nd power of generalized displacements, as given in the Appendix A. $\{\dots\}$ represents the terms of 3rd and higher powers of generalized displacements.

Without any application of distributed forces or moments, the internal forces ($\mathbf{t} = t_i \mathbf{e}_i$) and moments ($\mathbf{m} = m_i \mathbf{e}_i$) of cross section follow the static equilibrium given by

$$\frac{d\mathbf{t}}{ds} = 0 \quad \text{or} \quad \frac{dt_i}{ds} + \epsilon_{ijk} \hat{\kappa}_j t_k = 0 \quad (i = 1, 2, 3), \quad (7)$$

$$\frac{d\mathbf{m}}{ds} + \lambda \mathbf{e}_3 \times \mathbf{t} = 0 \quad \text{or} \quad \frac{dm_i}{ds} + \epsilon_{ijk} \hat{\kappa}_j m_k - \epsilon_{ij3} \lambda t_j = 0 \quad (i = 1, 2, 3), \quad (8)$$

in which ϵ_{ijk} is components of Eddington tensor.

For linear elastic and isotropic materials, the constitutive equations read

$$t_3 = EA(\lambda - 1), \quad m_1 = EI_1(\hat{\kappa}_1 - K_1), \quad m_2 = EI_2 \hat{\kappa}_2, \quad m_3 = GJ \hat{\kappa}_3, \quad (9)$$

where EA is the tensile stiffness, EI_i ($i = 1, 2$) are the bending stiffnesses, and GJ is the torsional stiffness. For beams with narrow rectangular cross-sections as illustrated in Fig. 2, the torsional stiffness GJ equals to $2EI_2/(1 + \nu)$, in which ν is the Poisson's ratio.

2.2. Double perturbation method

In general, the closed-form analytic solutions to the above nonlinear equations are unobtainable in most conditions. To bypass direct solutions, we introduce two small perturbation parameters, a and b , to expand the governing equations sequentially, and obtain the asymptotic solutions explicitly. The small parameter a is related to the deformation, assuming relative 'small' deformations from the onset of buckling. Quantitative calculations indicate that this method gives accurate predictions of the postbuckling deformations for applied strains up to ~40%, to be shown in the next section. The small parameter b is associated with the flexing level of initial planar curved beam, to be illustrated in more detail in the subsequent discussions. Although the small parameter b suggests slightly curved beams, the following calculations show that this double perturbation method can apply to arc-shaped beams with the central angle up to 180°, corresponding to semi-circles.

For the lateral postbuckling of planar curved beams, the perturbation parameter a could be set as the ratio of maximum out-of-plane displacement U_1 to the total arc-length (i.e., L_S). Since the deformation prior to buckling is negligible during the postbuckling, the displacements U_i and twist angle ϕ can be expanded in terms of the parameter a as (Su et al., 2012)

$$\begin{aligned} U_1 &= aU_{1(1)} + O(a^3), \\ U_2 &= a^2 U_{2(2)} + O(a^4), \\ U_3 &= a^2 U_{3(2)} + O(a^4), \\ \phi &= a\phi_{(1)} + O(a^3), \end{aligned} \quad (10)$$

in which the subscripts (1) and (2) refer to the 1st and the 2nd order in terms of parameter a . Substitution of Eq. (10) into

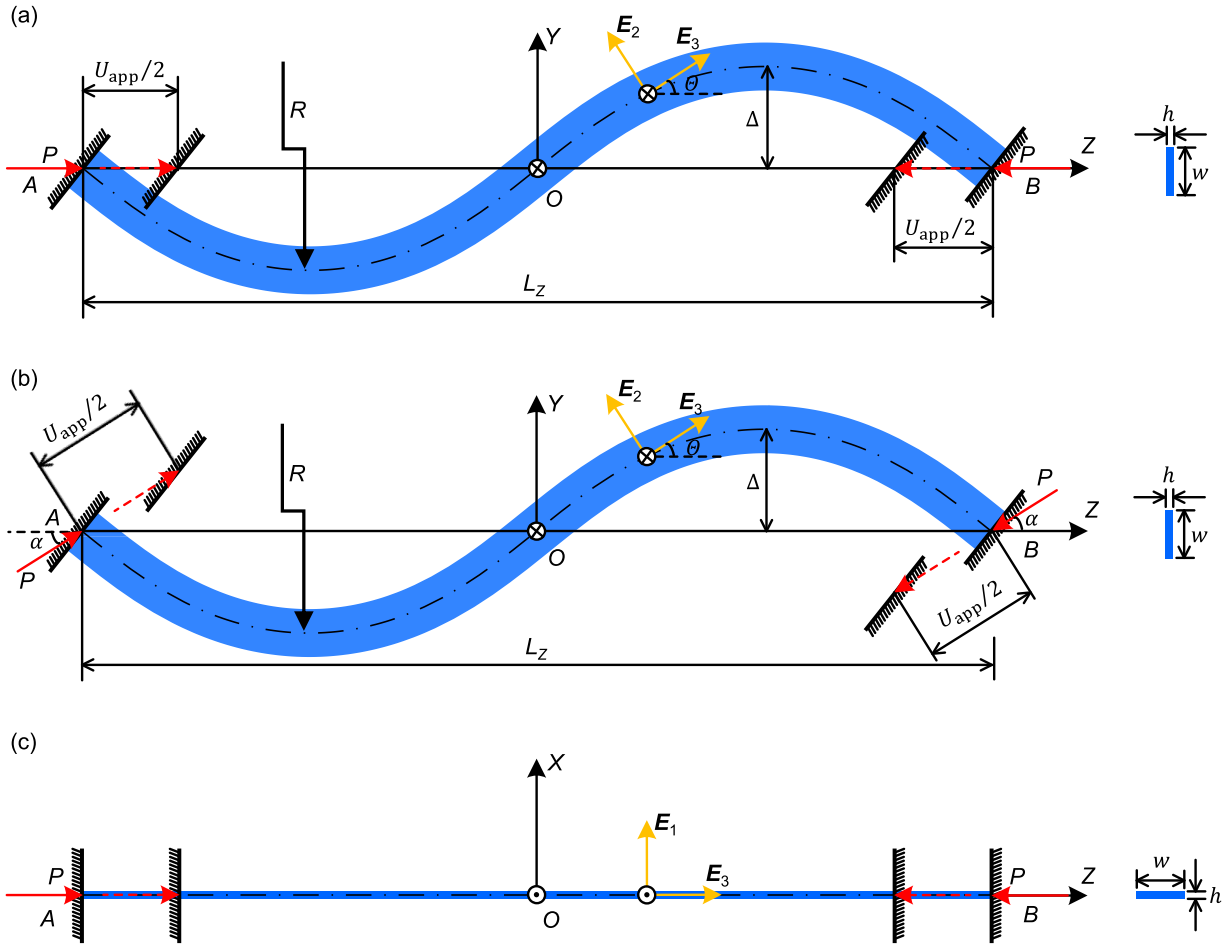


Fig. 2. Schematic illustration of the theoretical models for the end-to-end compression and the oblique compression. (a) and (b) Top views of a planar curved beam under an end-to-end compression and an oblique compression, respectively. (c) Front view of the above planar curved beam.

Eqs. (1), (5) and (6) gives the expansion forms of the elongation λ , the curvatures $\hat{\kappa}_i$, and the cosine coefficients a_{ij} ,

$$\begin{aligned}
 \lambda &= 1 + a^2 \lambda_{(2)} + O(a^4), \\
 \begin{Bmatrix} \hat{\kappa}_1 \\ \hat{\kappa}_2 \\ \hat{\kappa}_3 \end{Bmatrix} &= \begin{Bmatrix} K_1 \\ 0 \\ 0 \end{Bmatrix} + a \begin{Bmatrix} 0 \\ \hat{\kappa}_{2(1)} \\ \hat{\kappa}_{3(1)} \end{Bmatrix} + a^2 \begin{Bmatrix} \hat{\kappa}_{1(2)} \\ 0 \\ 0 \end{Bmatrix} + \begin{Bmatrix} O(a^4) \\ O(a^3) \\ O(a^3) \end{Bmatrix}, \\
 \{a_{ij}\} &= \begin{Bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{Bmatrix} + a \begin{Bmatrix} 0 & a_{12(1)} & a_{13(1)} \\ a_{21(1)} & 0 & 0 \\ a_{31(1)} & 0 & 0 \end{Bmatrix} \\
 &\quad + a^2 \begin{Bmatrix} a_{11(2)} & 0 & 0 \\ 0 & a_{22(2)} & a_{23(2)} \\ 0 & a_{32(2)} & a_{33(2)} \end{Bmatrix} + \begin{Bmatrix} O(a^4) & O(a^3) & O(a^3) \\ O(a^3) & O(a^4) & O(a^4) \\ O(a^3) & O(a^4) & O(a^4) \end{Bmatrix},
 \end{aligned} \tag{11}$$

in which

$$\lambda_{(2)} = U'_{3(2)} + U_{2(2)} K_1 + \frac{1}{2} U_{1(1)}^2, \tag{12a}$$

$$\begin{cases} \hat{\kappa}_{1(2)} = -U''_{2(2)} + (U_{3(2)} K_1)' + \psi_{(1)} U''_{1(1)} - \frac{1}{2} (\psi_{(1)}^2 + U_{1(1)}^2) K_1 \\ \hat{\kappa}_{2(1)} = U''_{1(1)} - \psi_{(1)} K_1 \\ \hat{\kappa}_{3(1)} = \phi'_{(1)} \end{cases}, \tag{12b}$$

$$\begin{cases} a_{11(2)} = -\frac{1}{2}(\psi_{(1)}^2 + U_{1(1)}'^2) \\ a_{12(1)} = -a_{21(1)} = \psi_{(1)} \\ a_{13(1)} = -a_{31(1)} = -U_{1(1)}' \\ a_{22(2)} = -\frac{1}{2}\psi_{(1)}^2 \\ a_{23(2)} = -U_{2(2)}' + U_{3(2)}K_1 + \psi_{(1)}U_{1(1)}' \\ a_{32(2)} = U_{2(2)}' - U_{3(2)}K_1 \\ a_{33(2)} = -\frac{1}{2}U_{1(1)}'^2 \end{cases}, \quad (12c)$$

in which $\psi_{(1)} = \int (\hat{\kappa}_{3(1)} - K_1 U_{1(1)}') dS = \phi_{(1)} - \int K_1 U_{1(1)}' dS$, and $()' = d()/dS$. The corresponding expansions of constitutive law Eq. (9) and the internal force/moments are given by

$$t_{3(n)} = EA\lambda_{(n)}, \quad m_{1(n)} = EI_1\hat{\kappa}_{1(n)}, \quad m_{2(n)} = EI_2\hat{\kappa}_{2(n)}, \quad m_{3(n)} = GJ\hat{\kappa}_{3(n)}, \quad (n = 1, 2, \dots), \quad (13)$$

$$\begin{aligned} t_3 &= t_{3(0)} + a^2 t_{3(2)} + O(a^4), \quad m_1 = m_{1(0)} + a^2 m_{1(2)} + O(a^4), \\ m_2 &= m_{2(0)} + a m_{2(1)} + O(a^3), \quad m_3 = m_{3(0)} + a m_{3(1)} + O(a^3). \end{aligned} \quad (14)$$

Analogously, the equilibrium equations (Eqs. (7) and (8)) are expanded as

$$\begin{aligned} \frac{dt_{i(n)}}{dS} + \sum_{l=0}^n (\epsilon_{ijk}\hat{\kappa}_{j(l)}t_{k(n-l)}) &= 0 \quad (i = 1, 2, 3; n = 0, 1, 2, \dots), \\ \frac{dm_{i(n)}}{dS} + \sum_{l=0}^n (\epsilon_{ijk}\hat{\kappa}_{j(l)}m_{k(n-l)} - \epsilon_{ij3}\lambda_{(l)}t_{j(n-l)}) &= 0 \quad (i = 1, 2, 3; n = 0, 1, 2, \dots), \end{aligned} \quad (15)$$

in which $\lambda_{(0)} = 1$, $\hat{\kappa}_{1(0)} = K_1$, and $\hat{\kappa}_{2(0)} = \hat{\kappa}_{3(0)} = 0$. Substitution of Eqs. (11) and (14) into Eq. (15) gives the expansions of the internal forces t_1 and t_2 as

$$t_1 = at_{1(1)} + O(a^3), \quad t_2 = t_{2(0)} + a^2 t_{2(2)} + O(a^4). \quad (16)$$

Note that the set of Eq. (15) for $n = 0$ denotes the equilibrium equations at the critical point, from which the internal forces and moments at the onset of buckling (i.e., $t_{2(0)}$, $t_{3(0)}$ and $m_{1(0)}$) can be determined. Alternatively, the Castigliano theorem can be also exploited to obtain these solutions at the onset of buckling, which are presented in Section 3.

After the expansion with respect to the parameter a , another expansion in terms of the perturbation parameter b is introduced for the above mechanics quantities (i.e., U_i , ϕ , λ , $\hat{\kappa}_i$, a_{ij} , t_i , m_i), which can be written in a unified manner as,

$$(\bullet)_{(n)} = (\bullet)_{(n)b(0)} + b^1(\bullet)_{(n)b(1)} + b^2(\bullet)_{(n)b(2)} + b^3(\bullet)_{(n)b(3)} + \dots \quad (n = 0, 1, 2, \dots), \quad (17)$$

where $(\bullet)$ refers to the above mechanics quantities; the subscript $(n)b(k)$ refers to the n th power of the parameter a and the k th power of the parameter b . Typically, the small parameter b can be set as

$$b = \frac{\Delta}{L_S} \text{ or } b = \frac{L_S}{R}, \quad (18)$$

where Δ is the maximum rise of the axis as shown in Fig. 2, and R is the minimum curvature radius. In the special condition of an arc-shaped beam, the curvature radius is a constant, and thereby, $b = L_S/R$ (i.e., the central angle) is more preferably used due to the simplified analyses. For the other beam shapes (e.g., sinusoidal-shaped beam), the definition $b = \Delta/L_S$ is more straightforward in the perturbation analyses. It is clear that the magnitude of b measures the flexing level of the initial configuration. For vanishing b , the curved beam degenerates into a straight one, which has been systematically studied by Su et al. (2012).

Generally, the coordinates (Y and Z) and the curvature K_1 of the central axis of the initial configuration (Fig. 2) can be written in form of the parameter functions of the arc length coordinate S ($\in [-L_S/2, L_S/2]$) as

$$Y = Y(S), \quad Z = Z(S) = \int_0^S \sqrt{1 - \left[\frac{dY(\xi)}{d\xi} \right]^2} d\xi, \quad K_1 = K_1(S) = \frac{d^2Z}{dS^2} \frac{dY}{dS} - \frac{d^2Y}{dS^2} \frac{dZ}{dS}. \quad (19)$$

Here, the validity of the above equation requires that the coordinate Y should be a single value for every prescribed coordinate Z . Otherwise, Eq. (19) should be given as piecewise expressions, which makes the following solutions more complicated. The current paper focuses on the former case in which Eq. (19) holds. In this condition, the oblique angle Θ that denotes the anticlockwise rotation from the Z axis to the local vector $\mathbf{E}_3$ is in the range of $[-\pi/2, \pi/2]$. This angle is related to the coordinates (Y and Z) by

$$\sin \Theta = Y'(S), \quad \cos \Theta = Z'(S). \quad (20)$$

Using the definition of the parameter b in Eq. (18), the coordinate $Y(S)$ can be written as

$$Y(S) = bL_S F(S, b) = \begin{cases} bL_S \left[\frac{1}{\Delta} Y(S) \right] & \text{for non-arc-shaped ribbons} \\ bL_S \left[\frac{1}{b^2} \left(\cos \frac{bS}{L_S} - \cos \frac{b}{2} \right) \right] & \text{for arc-shaped ribbons} \end{cases}, \quad (21)$$

in which $F(S, b)$ is a dimensionless shape function that characterizes the initial shape of the central axis.

Since $|\sin \Theta| = |Y'(S)| \leq 1$, the parameter b should satisfy the following inequality,

$$\max \left| bL_S \frac{\partial F(S, b)}{\partial S} \right| \leq 1. \quad (22)$$

Because the coordinate Y is single-valued for every given coordinate Z (see Eq. (19)), the above inequality requires that the upper limit $b_{\max}$ of parameter b corresponds to the smallest positive root of the equation $\max |bL_S \partial F(S, b) / \partial S| = 1$ ($S \in [-L_S/2, L_S/2]$). In other words, for a given shape function $F(S, b)$, the absolute value of the parameter b must be less than or equal to $b_{\max}$. For example, for the arc-shaped beam, the parameter b has to satisfy $b \leq b_{\max} = \pi$.

According to Eq. (21), the shape function $F(S, b)$ is an even function of the parameter b . With use of Eqs. (19) and (20), the expansions of the components related to the initial shape (e.g., $Y, Z, K_1, \sin \Theta, \cos \Theta$) in terms of the parameter b are given as

$$\begin{aligned} Y &= Y_{b(1)}b + Y_{b(3)}b^3 + O(b^5), \\ Z &= S + Z_{b(2)}b^2 + O(b^4), \\ K_1 &= K_{b(1)}b + K_{b(3)}b^3 + O(b^5), \\ \cos \Theta &= 1 + \cos \Theta_{b(2)}b^2 + O(b^4), \\ \sin \Theta &= \sin \Theta_{b(1)}b + \sin \Theta_{b(3)}b^3 + O(b^5). \end{aligned} \quad (23)$$

This set of equation shows that the geometric quantities related to the initial shape are either odd or even functions with respect to the parameter b .

Using Eq. (17), the geometric relations in Eq. (12) can be expanded in terms of the parameter b as

$$\lambda_{(2)b(k)} = U'_{3(2)b(k)} + \sum_{l=0}^k \left(U_{2(2)b(l)} K_{1b(k-l)} + \frac{1}{2} U'_{1(1)b(l)} U'_{1(1)b(k-l)} \right), \quad (24a)$$

$$\begin{cases} \hat{\kappa}_{1(2)b(k)} = -U''_{2(2)b(k)} + \sum_{l=0}^k \left[\left(U_{3(2)b(l)} K_{1b(k-l)} \right)' + \psi_{(1)b(l)} U''_{1(1)b(k-l)} \right] \\ \quad - \frac{1}{2} \sum_{l=0}^k \left\{ \left[\sum_{n=0}^l \left(\psi_{(1)b(n)} \psi_{(1)b(l-n)} + U'_{1(1)b(n)} U'_{1(1)b(l-n)} \right) \right] K_{1b(k-l)} \right\}, \\ \hat{\kappa}_{2(1)b(k)} = U''_{1(1)b(k)} - \sum_{l=0}^k \psi_{(1)b(l)} K_{1b(k-l)} \\ \hat{\kappa}_{3(1)b(k)} = \phi'_{(1)b(k)} \end{cases}, \quad (24b)$$

$$\begin{cases} a_{11(2)b(k)} = -\frac{1}{2} \sum_{l=0}^k \left(\psi_{(1)b(l)} \psi_{(1)b(k-l)} + U'_{1(1)b(l)} U'_{1(1)b(k-l)} \right) \\ a_{12(1)b(k)} = -a_{21(1)b(k)} = \psi_{(1)b(k)} \\ a_{13(1)b(k)} = -a_{31b(k)} = -U'_{1(1)b(k)} \\ a_{22(2)b(k)} = -\frac{1}{2} \sum_{l=0}^k \psi_{(1)b(l)} \psi_{(1)b(k-l)} \\ a_{23(2)b(k)} = -U'_{2(2)b(k)} + \sum_{l=0}^k \left(U_{3(2)b(l)} K_{1b(k-l)} + \psi_{(1)b(l)} U'_{1(1)b(k-l)} \right) \\ a_{32(2)b(k)} = U'_{2(2)b(k)} - \sum_{l=0}^k U_{3(2)b(l)} K_{1b(k-l)} \\ a_{33(2)b(k)} = -\frac{1}{2} \sum_{l=0}^k U'_{1(1)b(l)} U'_{1(1)b(k-l)} \end{cases}, \quad (24c)$$

in which $\psi_{(1)b(k)} = \int (\hat{\kappa}_{3(1)b(k)} - \sum_{l=0}^k K_{1b(l)} U'_{1(1)b(k-l)}) dS = \phi_{(1)b(k)} - \int \left(\sum_{l=0}^k K_{1b(l)} U'_{1(1)b(k-l)} \right) dS$.

The constitutive law Eq. (13) and the equilibrium equations Eq. (15) with $n = 1$ can be expanded as

$$\begin{aligned} t_{3(1)b(k)} &= EA \lambda_{(1)b(k)}, \quad m_{1(1)b(k)} = EI_1 \hat{\kappa}_{1(1)b(k)}, \quad m_{2(1)b(k)} = EI_2 \hat{\kappa}_{2(1)b(k)}, \\ m_{3(1)b(k)} &= GJ \hat{\kappa}_{3(1)b(k)} \quad (k = 0, 1, 2, \dots), \end{aligned} \quad (25)$$

$$\begin{aligned} t'_{1(1)b(k)} + \hat{\kappa}_{2(1)b(k)} t_{3(0)b(0)} - \hat{\kappa}_{3(1)b(k)} t_{2(0)b(0)} &= F_{I(1)b(k)}(S), \\ m'_{2(1)b(k)} + t_{1(1)b(k)} + \hat{\kappa}_{3(1)b(k)} m_{1(0)b(0)} &= F_{II(1)b(k)}(S), \\ m'_{3(1)b(k)} - \hat{\kappa}_{2(1)b(k)} m_{1(0)b(0)} &= F_{III(1)b(k)}(S), \end{aligned} \quad (26)$$

in which

$$\begin{aligned} F_{I(1)b(k)}(S) &= - \sum_{l=0}^k \left(\hat{\kappa}_{2(1)b(l)} t_{3(0)b(k-l)} - \hat{\kappa}_{3(1)b(l)} t_{2(0)b(k-l)} \right) + \hat{\kappa}_{2(1)b(k)} t_{3(0)b(0)} - \hat{\kappa}_{3(1)b(k)} t_{2(0)b(0)}, \\ F_{II(1)b(k)}(S) &= - \sum_{l=0}^k \left(\hat{\kappa}_{3(1)b(l)} m_{1(0)b(k-l)} - K_{1b(l)} m_{3(1)b(k-l)} \right) + \hat{\kappa}_{3(1)b(k)} m_{1(0)b(0)}, \\ F_{III(1)b(k)}(S) &= - \sum_{l=0}^k \left(K_{1b(l)} m_{2(1)b(k-l)} - \hat{\kappa}_{2(1)b(l)} m_{1(0)b(k-l)} \right) - \hat{\kappa}_{2(1)b(k)} m_{1(0)b(0)}. \end{aligned}$$

To facilitate the buckling process, the curved ribbons adopted in the 3D assembly techniques mostly have an ultrathin geometry, with the three characteristic dimensions $h \ll w \ll L_S$. As a result, the change of elongation (i.e., $\lambda - 1$) and the change of in-plane bending curvature (i.e., $\hat{\kappa}_1 - K_1$) during the flexural-torsional postbuckling are negligible as compared to the components related to out-of-plane bending and twisting. For simplicity, we neglect these terms, i.e., $\lambda - 1 = 0$, $\hat{\kappa}_1 - K_1 = 0$. Then by using Eq. (11), the expansion form of double perturbation for this assumption can be written as

$$\lambda_{(2)b(k)} = 0, \quad \hat{\kappa}_{1(2)b(k)} = 0. \quad (27)$$

For the sake of clarity, we adopt the symbol $a^n b^k$ to refer to the governing equations with the n th order of the parameter a and the k th order of the parameter b . The solution of the above governing equations can be then divided into two main steps:

Step 1: By solving the $a^1 b^k$ governing equations from $k = 0$ to $k = l$ sequentially, we can obtain each order of the critical load and buckling mode, e.g., the out-of-plane displacement $U_{1(1)b(k)}$ and the twist angle $\phi_{(1)b(k)}$ ($k = 0, \dots, l$).

Step 2: By solving the $a^2 b^k$ governing equations from $k = 0$ to $k = l$ sequentially, we can obtain each order of the applied displacement and the in-plane displacements (i.e., $U_{2(2)b(k)}$ and $U_{3(2)b(k)}$ ($k = 0, \dots, l$)).

In particular, the postbuckling behavior of a straight beam can be obtained by solving the $a^i b^0$ ($i = 1, 2$) governing equations. For a curved beam (i.e., $K_1 \neq 0$), the higher order terms of the parameter b should be taken into account. The detailed solution procedures for two representative loading conditions in buckling-guided 3D assembly are presented in the following subsections.

2.3. Buckling and postbuckling under an end-to-end compression

Consider a planar curved beam of a rectangular section ($w \times h$) subject to an end-to-end compression, as shown in Fig. 2(a) and (c). During the experiments of buckling-guided assembly, the two ends (i.e., $S = -L_S/2, L_S/2$) are typically bonded to an elastomer substrate, which can be simplified as clamped boundaries in the analyses. The resulting boundary conditions are given by

$$\begin{aligned} U_1|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} &= 0, \quad \mathbf{e}_i|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = \mathbf{E}_i|_{S=-\frac{L_S}{2}, \frac{L_S}{2}}, \\ [(U_2 \mathbf{E}_2 + U_3 \mathbf{E}_3) \bullet (\cos \Theta \mathbf{E}_2 + \sin \Theta \mathbf{E}_3)]|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} &= 0, \\ [\mathbf{t} \bullet (-\sin \Theta \mathbf{E}_2 + \cos \Theta \mathbf{E}_3)]|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} &= -P, \\ \frac{U_{\text{app}}}{2} &= [(U_2 \mathbf{E}_2 + U_3 \mathbf{E}_3) \bullet (-\sin \Theta \mathbf{E}_2 + \cos \Theta \mathbf{E}_3)]|_{S=-\frac{L_S}{2}} \\ &= -[(U_2 \mathbf{E}_2 + U_3 \mathbf{E}_3) \bullet (-\sin \Theta \mathbf{E}_2 + \cos \Theta \mathbf{E}_3)]|_{S=\frac{L_S}{2}}, \end{aligned} \quad (28)$$

where P and U_{app} are the applied force and the corresponding applied displacement respectively, which can be expanded as:

$$\begin{aligned} U_{\text{app}} &= a^2 U_{\text{app}(2)} + O(a^4), \\ U_{\text{app}(2)} &= U_{\text{app}(2)b(0)} + b U_{\text{app}(2)b(1)} + b^2 U_{\text{app}(2)b(2)} + O(b^3), \end{aligned} \quad (29)$$

$$\begin{aligned} P &= P_{(0)} + a^2 P_{(2)} + O(a^4), \\ P_{(0)} &= P_{(0)b(0)} + b P_{(0)b(1)} + b^2 P_{(0)b(2)} + O(b^3), \\ P_{(2)} &= P_{(2)b(0)} + b P_{(2)b(1)} + b^2 P_{(2)b(2)} + O(b^3). \end{aligned} \quad (30)$$

Using the double perturbation method and Eqs. (4) and (6), the boundary condition Eq. (28) can be rewritten as

$$U_{1(1)b(k)}|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \quad \psi_{(1)b(k)}|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \quad U'_{1(1)b(k)}|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \quad (31)$$

$$\begin{aligned}
& \sum_{l=0}^k \left[\cos \Theta_{b(l)} U_{2(2)b(k-l)} + \sin \Theta_{b(l)} U_{3(2)b(k-l)} \right] \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \\
& \sum_{l=0}^k \left[-\sin \Theta_{b(l)} t_{2(2)b(k-l)} + \cos \Theta_{b(l)} t_{3(2)b(k-l)} \right] \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = -P_{(2)b(k)}, \\
& \left(U'_{2(2)b(k)} - \sum_{l=0}^k U_{3(2)b(l)} K_{1b(k-l)} \right) \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \\
& \frac{U_{app(2)b(k)}}{2} = \sum_{l=0}^k \left[-\sin \Theta_{b(l)} U_{2(2)b(k-l)} + \cos \Theta_{b(l)} U_{3(2)b(k-l)} \right] \Big|_{S=-\frac{L_S}{2}} \\
& \quad = - \sum_{l=0}^k \left[-\sin \Theta_{b(l)} U_{2(2)b(k-l)} + \cos \Theta_{b(l)} U_{3(2)b(k-l)} \right] \Big|_{S=\frac{L_S}{2}}.
\end{aligned} \tag{32}$$

Before solving the a^{1bk} governing equations, the internal forces (i.e., $t_{i(0)}$) and internal moments (i.e., $m_{i(0)}$) at the onset of buckling should be determined first. Prior to the lateral buckling, the internal force $t_{1(0)}$ and the internal moments $m_{2(0)}$ and $m_{3(0)}$ are zero, and the other three components (i.e., $t_{2(0)}$, $t_{3(0)}$, $m_{1(0)}$) can be obtained by the Castigliano theorem. Here the terms up to the 1st order with respect to the parameter b are given as

$$t_{2(0)b(0)} = 0, \quad t_{3(0)b(0)} = -P_{(0)b(0)}, \quad m_{1(0)b(0)} = 0, \tag{33}$$

$$\begin{aligned}
t_{2(0)b(1)} &= P_{(0)b(0)} \left[-\frac{12}{L_S^3} \int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} S Y_{b(1)}(S) dS + \sin \Theta_{b(1)} \right], \\
t_{3(0)b(1)} &= -P_{(0)b(1)}, \\
m_{1(0)b(1)} &= P_{(0)b(0)} \left[-\frac{1}{L_S} \int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} Y_{b(1)}(S) dS - \frac{12S}{L_S^3} \int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} S Y_{b(1)}(S) dS + Y_{b(1)}(S) \right].
\end{aligned} \tag{34}$$

Substitution of Eqs. (25) and (33) into the equilibrium equation Eq. (26) gives the differential equation for $\hat{\kappa}_{2(1)b(k)}$,

$$\hat{\kappa}_{2(1)b(k)}'' + \omega_{(0)b(0)}^2 \hat{\kappa}_{2(1)b(k)} = \frac{1}{EI_2} [F'_{II(1)b(k)}(S) - F_{I(1)b(k)}(S)], \tag{35}$$

in which $\omega_{(0)b(0)}^2 = P_{(0)b(0)}/(EI_2)$. The solution of the above equation is

$$\begin{aligned}
\hat{\kappa}_{2(1)b(k)} &= C_{(1)b(k)(1)} \cos(\omega_{(0)b(0)} S) + C_{(1)b(k)(2)} \sin(\omega_{(0)b(0)} S) \\
&+ \int_0^S \frac{\sin[\omega_{(0)b(0)}(S - \xi)] [F'_{II(1)b(k)}(\xi) - F_{I(1)b(k)}(\xi)]}{EI_2 \omega_{(0)b(0)}} d\xi,
\end{aligned} \tag{36}$$

where $C_{(1)b(k)(1)}$ and $C_{(1)b(k)(2)}$ are unknown parameters to be determined by boundary conditions. Using Eqs. (3), (12) and (26), the twisting curvature $\hat{\kappa}_{3(1)b(k)}$, internal force $t_{1(1)b(k)}$, component $\psi_{(1)b(k)}$, twist angle $\phi_{(1)b(k)}$ and out-of-plane displacement $U_{1(1)b(k)}$ can be derived as

$$\begin{aligned}
t_{1(1)b(k)} &= F_{II(1)b(k)}(S) - EI_2 \hat{\kappa}_{2(1)b(k)}', \\
\hat{\kappa}_{3(1)b(k)} &= \frac{m_{3(1)b(k)}}{GJ} = \frac{1}{GJ} \int_0^S F_{III(1)b(k)}(\xi) d\xi + C_{(1)b(k)(3)},
\end{aligned} \tag{37}$$

$$\psi_{(1)b(k)} = \int_0^S \left(\hat{\kappa}_{3(1)b(k)}(\xi) - \sum_{l=0}^k K_{1b(l)}(\xi) U'_{1(1)b(k-l)}(\xi) \right) d\xi + C_{(1)b(k)(4)}, \tag{38}$$

$$\begin{aligned}
\phi_{(1)b(k)} &= \int \hat{\kappa}_{3(1)b(k)}(\xi) d\xi, \\
U_{1(1)b(k)} &= \int_0^S \int_0^\xi \left(\hat{\kappa}_{2(1)b(k)}(\zeta) + \sum_{l=0}^k \psi_{(1)b(l)}(\zeta) K_{1b(k-l)}(\zeta) \right) d\zeta d\xi \\
&+ C_{(1)b(k)(5)} S + C_{(1)b(k)(6)},
\end{aligned} \tag{39}$$

in which $C_{(1)b(k)(i)}$ ($i = 3, 4, 5, 6$) are the other four parameters to be determined. Substitution of Eqs. (38) and (39) into the boundary Eq. (31) gives a system of linear equations for the unknown parameters $C_{(1)b(k)(i)}$ ($i = 1, 2, \dots, 6$). For $k = 0$, the linear equations are homogeneous, and the existence of a non-zero solution gives an algebraic equation whose smallest

positive root is the critical load $P_{(0)b(0)}$. By introducing an additional equation (i.e., the maximum value of $U_{1(1)b(0)}$ is L_S), the parameters $C_{(1)b(0)(i)}$ ($i = 1, 2, \dots, 6$) can be fully determined. For $k = 1, 2, \dots$, the linear equations are non-homogeneous, and the existence of a solution gives the higher order terms of the critical load (i.e., $P_{(0)b(k)}$ ($k = 1, 2, \dots$)). Similarly, by solving these equations with the orthogonality condition (van der Heijden, 2009), $\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} U_{1(1)b(k)}(S)U_{1(1)b(0)}(S)dS = 0$, we can determine the higher power components of the parameters, i.e., $C_{(1)b(k)(i)}$ ($k = 1, 2, 3, \dots$; $i = 1, 2, \dots, 6$). Then, the buckling mode in terms of $\phi_{(1)b(k)}$ and $U_{1(1)b(k)}$ can be obtained by substituting these parameters $C_{(1)b(k)(i)}$ into Eq. (39).

For the $a^2 b^k$ governing equations, substitution of geometric relation Eq. (12) into Eq. (27) gives the expressions of the in-plane displacements $U_{2(2)b(k)}$ and $U_{3(2)b(k)}$,

$$\begin{aligned} U_{2(2)b(k)} &= \int_0^S \int_0^\xi \left\{ \sum_{l=0}^k \left[(U_{3(2)b(l)}(\xi) K_{1b(k-l)}(\xi))' + \psi_{1b(l)}(\xi) U_{1(1)b(k-l)}''(\xi) \right] \right. \\ &\quad \left. - \frac{1}{2} \sum_{l=0}^k \left\{ \sum_{n=0}^l (\psi_{1b(n)}(\xi) \psi_{1b(l-n)}(\xi) + U_{1(1)b(n)}'(\xi) U_{1(1)b(l-n)}'(\xi)) \right\} K_{1b(k-l)}(\xi) \right\} d\xi d\xi \\ &\quad + C_{(2)b(k)(1)} S + C_{(2)b(k)(2)}, \\ U_{3(2)b(k)} &= \int_0^S \left[- \sum_{l=0}^k (U_{2(2)b(l)}(\xi) K_{1b(k-l)}(\xi) + \frac{1}{2} U_{1(1)b(l)}'(\xi) U_{1(1)b(k-l)}'(\xi)) \right] d\xi + C_{(2)b(k)(3)}, \end{aligned} \quad (40)$$

in which $C_{(2)b(k)(i)}$ ($i = 1, 2, 3$) are unknown parameters.

To this end, the parameters $C_{(2)b(k)(i)}$ ($i = 1, 2, 3$) can be determined by three dominant boundary conditions related to the constraint of in-plane rigid-body motions, including a zero transverse displacement (i.e., normal direction to the compression) and the same level of applied displacement at the two ends. These boundary equations are extracted from Eq. (32) as

$$\begin{aligned} \sum_{l=0}^k [\cos \Theta_{b(l)} U_{2(2)b(k-l)} + \sin \Theta_{b(l)} U_{3(2)b(k-l)}] \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} &= 0, \\ \sum_{l=0}^k [-\sin \Theta_{b(l)} U_{2(2)b(k-l)} + \cos \Theta_{b(l)} U_{3(2)b(k-l)}] \Big|_{S=-\frac{L_S}{2}} &= - \sum_{l=0}^k [-\sin \Theta_{b(l)} U_{2(2)b(k-l)} + \cos \Theta_{b(l)} U_{3(2)b(k-l)}] \Big|_{S=\frac{L_S}{2}}. \end{aligned} \quad (41)$$

Then the parameters $C_{(2)b(k)(i)}$ ($i = 1, 2, 3$) and the in-plane displacements ($U_{2(2)b(k)}$ and $U_{3(2)b(k)}$) can be obtained by substitution of Eq. (40) into Eq. (41). After determining the in-plane displacements, the applied displacement loading $U_{app(2)b(k)}$ can be obtained as

$$\begin{aligned} U_{app(2)b(k)} &= 2 \sum_{l=0}^k [-\sin \Theta_{b(l)} U_{2(2)b(k-l)} + \cos \Theta_{b(l)} U_{3(2)b(k-l)}] \Big|_{S=-\frac{L_S}{2}} \\ &= -2 \sum_{l=0}^k [-\sin \Theta_{b(l)} U_{2(2)b(k-l)} + \cos \Theta_{b(l)} U_{3(2)b(k-l)}] \Big|_{S=\frac{L_S}{2}}. \end{aligned} \quad (42)$$

Together with the components (e.g., $U_{1(1)b(k)}$, $\phi_{(1)b(k)}$) obtained from the $a^1 b^k$ governing equations, the deformed configuration during postbuckling can be determined.

For a certain initial configuration with prescribed shape function $F(S, b)$ and value of parameter b , the perturbation parameter a can be calculated from

$$a = \pm \sqrt{\frac{U_{app}}{\sum_{k=0}^l b^k U_{app(2)b(k)}}}, \quad (43)$$

for displacement (U_{app}) loading, where the symbol $\pm$ depends on the direction of the out-of-plane displacement, and the integer l denotes the order of terms adopted in the perturbation method. By substituting the value of parameter a into the expansion forms of the generalized displacement components (i.e., U_i , ϕ), the deformed configuration during postbuckling is then determined.

2.4. Buckling and postbuckling under an oblique compression

During the mechanically-guided 3D assembly, the 2D ribbon-shaped precursor often experiences an oblique compression when the end-to-end line deviates from the direction of uniaxial prestretch, as shown in Fig. 1(a). The corresponding mechanics model is illustrated in Fig. 2(b) and (c), with α denoting the loading angle. Here, the boundary conditions are given by

$$U_{1(1)b(k)} \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \quad \psi_{1b(k)} \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \quad U_{1(1)b(k)}' \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \quad (44)$$

$$\begin{aligned}
& \sum_{l=0}^k \left[\begin{aligned} & (\cos \alpha \cos \Theta_{b(l)} + \sin \alpha \sin \Theta_{b(l)}) U_{2(2)b(k-l)} \\ & + (\cos \alpha \sin \Theta_{b(l)} - \sin \alpha \cos \Theta_{b(l)}) U_{3(2)b(k-l)} \end{aligned} \right] \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \\
& \sum_{l=0}^k \left[\begin{aligned} & (-\cos \alpha \sin \Theta_{b(l)} + \sin \alpha \cos \Theta_{b(l)}) t_{2(2)b(k-l)} \\ & + (\cos \alpha \cos \Theta_{b(l)} + \sin \alpha \sin \Theta_{b(l)}) t_{3(2)b(k-l)} \end{aligned} \right] \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = -P_{(2)b(k)}, \\
& \left(U'_{2(2)b(k)} - \sum_{l=0}^k U_{3(2)b(l)} K_{1b(k-l)} \right) \Big|_{S=-\frac{L_S}{2}, \frac{L_S}{2}} = 0, \\
& \frac{U_{\text{app}(2)b(k)}}{2} = \sum_{l=0}^k \left[\begin{aligned} & (-\cos \alpha \sin \Theta_{b(l)} + \sin \alpha \cos \Theta_{b(l)}) U_{2(2)b(k-l)} \\ & + (\cos \alpha \cos \Theta_{b(l)} + \sin \alpha \sin \Theta_{b(l)}) U_{3(2)b(k-l)} \end{aligned} \right] \Big|_{S=-\frac{L_S}{2}} \\
& = - \sum_{l=0}^k \left[\begin{aligned} & (-\cos \alpha \sin \Theta_{b(l)} + \sin \alpha \cos \Theta_{b(l)}) U_{2(2)b(k-l)} \\ & + (\cos \alpha \cos \Theta_{b(l)} + \sin \alpha \sin \Theta_{b(l)}) U_{3(2)b(k-l)} \end{aligned} \right] \Big|_{S=\frac{L_S}{2}}.
\end{aligned} \tag{45}$$

Similar to the case of end-to-end compression, the first term of the expansions with respect to the parameter b for the internal forces and the internal moment at the onset of buckling are obtained using the Castigliano theorem, i.e.,

$$\begin{aligned}
t_{2(0)b(0)} &= \frac{24I_1 P_{(0)b(0)} \sin \alpha}{-12I_1 - AL_S^2 + (12I_1 - AL_S^2) \cos(2\alpha)}, \\
t_{3(0)b(0)} &= -\frac{2AL_S^2 P_{(0)b(0)} \cos \alpha}{12I_1 + AL_S^2 + (-12I_1 + AL_S^2) \cos(2\alpha)}, \\
m_{1(0)b(0)} &= \frac{24I_1 P_{(0)b(0)} \sin \alpha}{-12I_1 - AL_S^2 + (12I_1 - AL_S^2) \cos(2\alpha)} S.
\end{aligned} \tag{46}$$

Substitution of the constitutive law Eq. (25) into the equilibrium Eq. (26) gives a more complicated differential equation than the case of end-to-end compression:

$$\begin{aligned}
& \frac{d}{dS} \left(\frac{d^2 \hat{\kappa}_{3(1)b(k)}}{dS^2} - \frac{2}{S} \frac{d \hat{\kappa}_{3(1)b(k)}}{dS} + (\beta^4 S^2 + \gamma \beta^2) \hat{\kappa}_{3(1)b(k)} \right) = -\frac{t_{3(0)b(0)} F_{\text{III}(1)b(k)}(S)}{GJ E I_2} - \frac{m_{1(0)b(0)} F_{\text{I}(1)b(k)}(S)}{GJ E I_2} \\
& + \frac{m_{1(0)b(0)}}{GJ} \frac{d^2}{dS^2} \left[\frac{F_{\text{III}(1)b(k)}(S)}{m_{1(0)b(0)}} \right] + \frac{m_{1(0)b(0)}}{GJ E I_2} \frac{d}{dS} [F_{\text{II}(1)b(k)}(S)],
\end{aligned} \tag{47}$$

in which $\beta = \sqrt{|m_{1(0)b(0)}|/S} / \sqrt{GJ E I_2}$, $\gamma = -t_{3(0)b(0)}/(E I_2 \beta^2)$.

The solution of Eq. (47) is

$$\begin{aligned}
\hat{\kappa}_{3(1)b(k)}(S) &= C_{(1)b(k)(1)} T_{\text{I}(0)b(0)}(\beta S) + C_{(1)b(k)(2)} T_{\text{II}(0)b(0)}(\beta S) \\
&+ C_{(1)b(k)(3)} T_{\text{III}(0)b(0)}(\beta S) + T_{\text{IV}(1)b(k)}(\beta S),
\end{aligned} \tag{48}$$

where $C_{(1)b(k)(i)}$ ($i = 1, 2, 3$) are the parameters to be determined, and the expressions of $T_{\text{I}(0)b(0)}$, $T_{\text{II}(0)b(0)}$, $T_{\text{III}(0)b(0)}$, and $T_{\text{IV}(1)b(k)}$ are given in Supporting Information. Then the bending curvature $\hat{\kappa}_{2(1)b(k)}$, internal force $t_{1(1)b(k)}$, component $\psi_{(1)b(k)}$, twist angle $\phi_{(1)b(k)}$, and out-of-plane displacement $U_{1(1)b(k)}$ can be derived as

$$\begin{aligned}
\hat{\kappa}_{2(1)b(k)} &= \frac{m_{2(1)b(k)}}{E I_2} = \frac{GJ \hat{\kappa}'_{3(1)b(k)} - F_{\text{III}(1)b(k)}(S)}{m_{1(0)b(0)}}, \\
t_{1(1)b(k)} &= F_{\text{II}(1)b(k)}(S) - E I_2 \hat{\kappa}'_{2(1)b(k)} - \hat{\kappa}_{3(1)b(k)} m_{1(0)b(0)},
\end{aligned} \tag{49}$$

$$\psi_{(1)b(k)} = \int_0^S \left(\hat{\kappa}_{3(1)b(k)}(\xi) - \sum_{l=0}^k K_{1b(l)}(\xi) U'_{1(1)b(k-l)}(\xi) \right) d\xi + C_{(1)b(k)(4)}, \tag{50}$$

$$\begin{aligned}
\phi_{(1)b(k)} &= \int \hat{\kappa}_{3(1)b(k)}(\xi) d\xi, \\
U_{1(1)b(k)} &= \int_0^S \int_0^\xi \left(\hat{\kappa}_{2(1)b(k)}(\zeta) + \sum_{l=0}^k \psi_{(1)b(l)}(\zeta) K_{1b(k-l)}(\zeta) \right) d\zeta d\xi \\
&+ C_{(1)b(k)(5)} S + C_{(1)b(k)(6)},
\end{aligned} \tag{51}$$

where $C_{(1)b(k)(i)}$ ($i = 4, 5, 6$) are unknown parameters that can be determined by the boundary conditions and normalization condition. The subsequent procedures are the same as the case of end-to-end compression in Section 2.3, and are not repeated here. Note that the above theory can degenerate into that of straight beams developed by Su et al. (2012), for vanishing b .

3. Results and discussion

3.1. Analytic solutions of displacements and curvatures for general ribbon shapes

According to the solution procedures described in the [Sections 2.3](#), the analytic solutions of displacements and curvatures during postbuckling of a general planar ribbon can be obtained for the end-to-end compression. For the sake of brevity, only the associated expressions up to a^2b^1 order are provided, as given by

$$P_{(0)b(0)} = \frac{4\pi^2 EI_2}{L_S^2}, \quad P_{(0)b(1)} = 0, \quad (52)$$

$$U_{\text{app}(2)b(0)} = \frac{\pi^2 L_S}{4}, \quad U_{\text{app}(2)b(1)} = 0, \quad (53)$$

$$U_{1(1)b(0)} = L_S \cos^2 \frac{\pi S}{L_S}, \quad U_{1(1)b(1)} = 0, \\ U_{2(2)b(0)}(S) = 0,$$

$$U_{2(2)b(1)}(S) = \frac{\pi}{32L_S^2} \left\langle \begin{aligned} &(-L_S + 2S) \int_0^{-\frac{L_S}{2}} \int_0^\xi \tilde{F}_2(\zeta) d\zeta d\xi - (L_S + 2S) \int_0^{\frac{L_S}{2}} \int_0^\xi \tilde{F}_2(\zeta) d\zeta d\xi \\ &+ 2L_S \left\{ \int_0^S \int_0^\xi \tilde{F}_2(\zeta) d\zeta d\xi + \pi L_S \left[\begin{aligned} &(-L_S + 2S) \sin \Theta_{b(1)} \Big|_{S=-\frac{L_S}{2}} \\ &+ (L_S + 2S) \sin \Theta_{b(1)} \Big|_{S=\frac{L_S}{2}} \end{aligned} \right] \right\} \end{aligned} \right\rangle, \quad (54)$$

$$U_{3(2)b(0)}(S) = -\frac{\pi}{16} \left(4\pi S - L_S \sin \frac{4\pi S}{L_S} \right), \quad U_{3(2)b(1)}(S) = 0,$$

$$\phi_{(1)b(0)} = 0, \quad \phi_{(1)b(1)} = \int \left(\frac{1}{GJ} \int_0^S F_{\text{III}(1)b(1)}(\eta) d\eta - \frac{1}{L_S} \int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} \tilde{F}_1(\xi) d\xi \right) dS,$$

$$\hat{\kappa}_{2(1)b(0)} = -\frac{2\pi^2}{L_S} \cos \left(\frac{2\pi S}{L_S} \right), \quad \hat{\kappa}_{2(1)b(1)} = 0, \quad (55)$$

$$\hat{\kappa}_{3(1)b(0)} = 0, \quad \hat{\kappa}_{3(1)b(1)} = \frac{1}{GJ} \int_0^S F_{\text{III}(1)b(1)}(\eta) d\eta - \frac{1}{L_S} \int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} \tilde{F}_1(\xi) d\xi,$$

in which,

$$F_{\text{III}(1)b(1)}(S) = \frac{2\pi^2}{L_S} \cos \frac{2\pi S}{L_S} [EI_2 K_{1b(1)}(S) - m_{1(0)b(1)}(S)], \\ \tilde{F}_1(S) = \frac{1}{GJ} \int_0^S F_{\text{III}(1)b(1)}(\eta) d\eta + \pi K_{1b(1)}(S) \sin \frac{2\pi S}{L_S}, \\ \tilde{F}_2(S) = 8\pi L_S \left(\cos \frac{4\pi S}{L_S} - 1 \right) K_{1b(1)}(S) + L_S \left[-4\pi S + L_S \sin \frac{4\pi S}{L_S} \right] K'_{1b(1)}(S) \\ + 16\pi \cos \frac{2\pi S}{L_S} \left(\int_S^{-\frac{L_S}{2}} \tilde{F}_1(\xi) d\xi + \int_S^{\frac{L_S}{2}} \tilde{F}_1(\xi) d\xi + \frac{2S}{L_S} \int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} \tilde{F}_1(\xi) d\xi \right).$$

The above equations show that the displacement and curvature components are either odd or even functions of the parameter b , when expanded to a^2b^1 order. This can be attributed to the oddity of the geometric quantities related to the initial shape ([Eq. \(23\)](#)). Furthermore, our analytic results for expansion to a higher order of the parameter b indicate that the above mechanics quantities can be expanded in the following forms,

$$P = (P_{(0)b(0)} + P_{(0)b(2)}b^2 + \dots) + \dots, \quad (56)$$

$$U_{\text{app}} = (U_{\text{app}(2)b(0)} + U_{\text{app}(2)b(2)}b^2 + \dots)a^2 + \dots, \quad (57)$$

$$U_1 = (U_{1(1)b(0)} + U_{1(1)b(2)}b^2 + \dots)a + \dots, \\ U_2 = (U_{2(2)b(1)}b + U_{2(2)b(3)}b^3 + \dots)a^2 + \dots, \\ U_3 = (U_{3(2)b(0)} + U_{3(2)b(2)}b^2 + \dots)a^2 + \dots, \\ \phi = (\phi_{(1)b(1)}b + \phi_{(1)b(3)}b^3 + \dots)a + \dots, \quad (58)$$

$$\hat{\kappa}_2 = (\hat{\kappa}_{2(1)b(0)} + \hat{\kappa}_{2(1)b(2)}b^2 + \dots)a + \dots, \\ \hat{\kappa}_3 = (\hat{\kappa}_{3(1)b(1)}b + \hat{\kappa}_{3(1)b(3)}b^3 + \dots)a + \dots. \quad (59)$$

The peak value ε_{max} of maximum principal strain in the ribbon is given as ([Fan et al., 2017](#))

$$\varepsilon_{\text{max}} = ah \max \left(\frac{1-\nu}{4} |\hat{\kappa}_{2(1)}| + \frac{1}{2} \sqrt{\frac{(1+\nu)^2}{4} \hat{\kappa}_{2(1)}^2 + \hat{\kappa}_{3(1)}^2} \right). \quad (60)$$

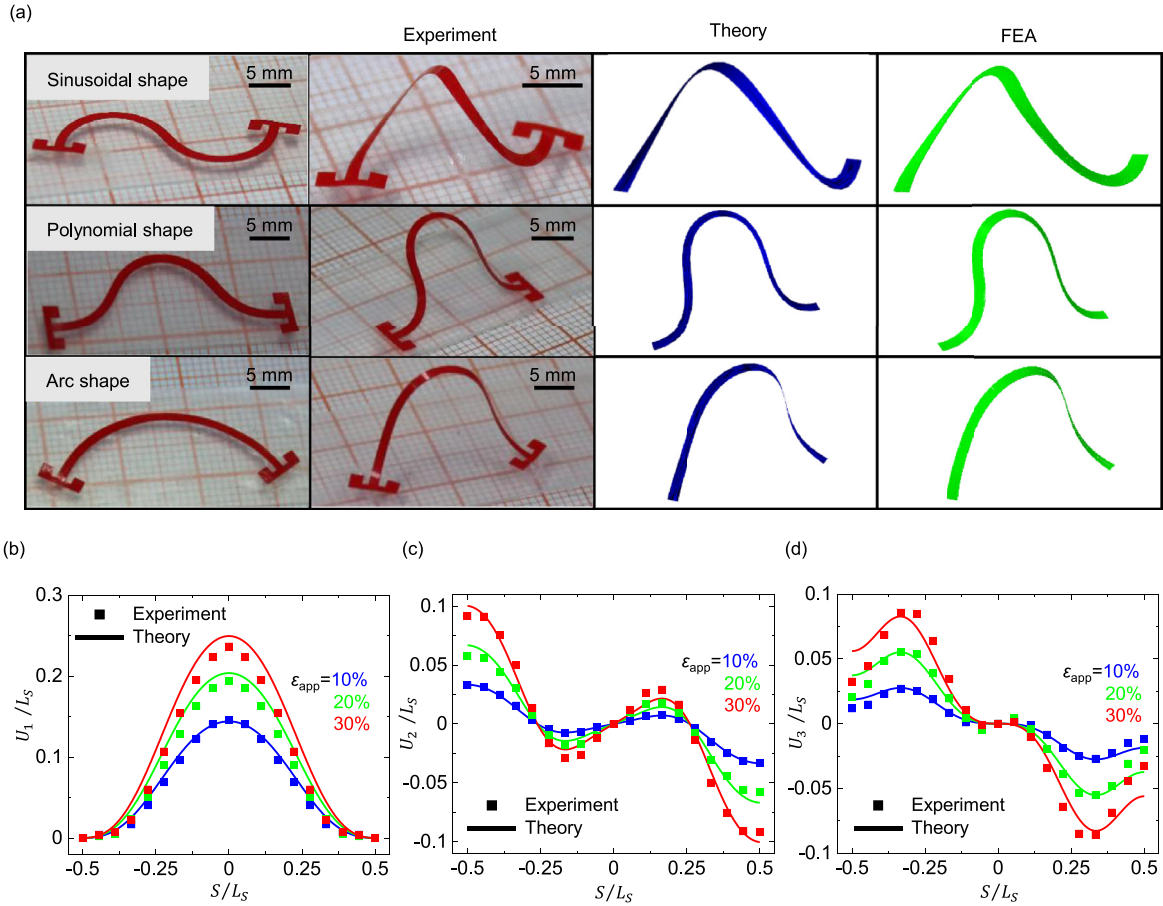


Fig. 3. (a) Experimental, theoretical and FEA results on the deformed configurations of three different ribbons ($b = 0.15, 0.25$ and $2\pi/3$ for the sinusoidal, polynomial, and arc shapes) under end-to-end compressions ($\epsilon_{app} = 30\%$). (b)–(d) Experimental and theoretical results on the distribution of dimensionless displacement components for the sinusoidal-shaped ribbon ($b = 0.15$) under three different levels of compression ($\epsilon_{app} = 10\%, 20\%$ and 30%).

The applied strain ϵ_{app} is defined as

$$\epsilon_{app} = \frac{U_{app}}{L_Z} = \frac{(U_{app(2)b(0)} + U_{app(2)b(2)}b^2 + \dots)a^2 + \dots}{L_Z}, \quad (61)$$

where L_Z is the initial span of the beam as illustrated in Fig. 2. The scaling analyses on Eqs. (58)–(60) with the aid of Eq. (61) yield

$$U_1 \sim \sqrt{\epsilon_{app}}b^0, \quad U_2 \sim \epsilon_{app}b, \quad U_3 \sim \epsilon_{app}b^0, \quad \phi \sim \sqrt{\epsilon_{app}}b^1, \quad (62)$$

$$\hat{\kappa}_2 \sim \sqrt{\epsilon_{app}}b^0, \quad \hat{\kappa}_3 \sim \sqrt{\epsilon_{app}}b^1, \quad (63)$$

$$\epsilon_{max} \sim \sqrt{\epsilon_{app}}b^0. \quad (64)$$

To quantify the contribution of twisting deformations during the postbuckling, a mode ratio R_{mode} (Xu et al., 2015) was introduced, which can be derived as

$$R_{mode} = \frac{\int_{-L_S/2}^{L_S/2} |\hat{\kappa}_3(S)| dS}{\int_{-L_S/2}^{L_S/2} |\hat{\kappa}_2(S)| dS} \approx b \frac{\int_{-L_S/2}^{L_S/2} |\hat{\kappa}_{3(1)b(1)}(S)| dS}{\int_{-L_S/2}^{L_S/2} |\hat{\kappa}_{2(1)b(0)}(S)| dS} \sim b. \quad (65)$$

This equation suggests that the mode ratio R_{mode} is proportional to the parameter b , if only the primary term is accounted for. It also reveals that the mode ratio R_{mode} is independent on the applied strain ϵ_{app} , and is affected mainly by the initial shape of central axis and the material constant (i.e., Poisson's ratio).

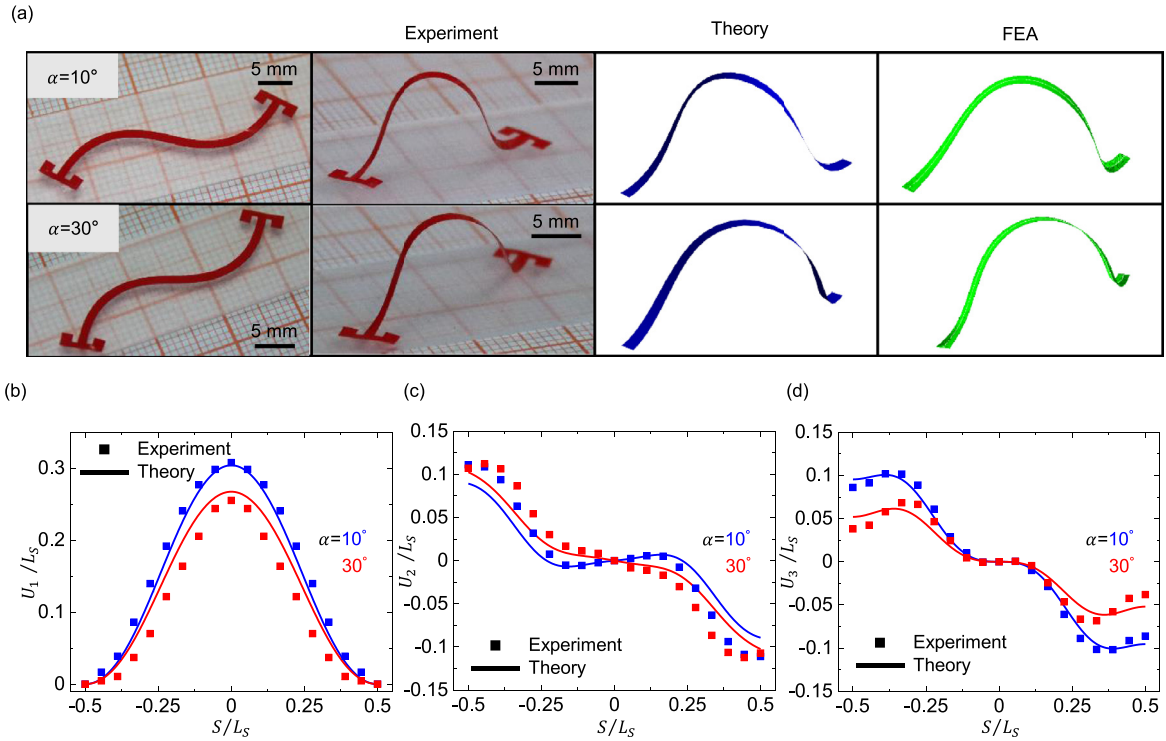


Fig. 4. (a) Experimental, theoretical and FEA results on the deformed configurations of sinusoidal-shaped ribbon ($b = 0.1$) under oblique compressions ($\varepsilon_{app} = 30\%$; loading angle $\alpha = 10^\circ$ and 30°). (b)–(d) Experimental and theoretical results on the distribution of the dimensionless displacement components for the compressed sinusoidal-shaped ribbon in (a).

For the oblique compression, the solutions of deformation components can be obtained in a similar manner, but they do not exhibit the above odd or even characteristics with respect to the parameter b . These results are not provided here due to their complexity. Note that the applied strain is given by $\varepsilon_{app} = U_{app}/(L_Z \cos \alpha)$ in this condition.

3.2. Analytic solutions of displacements and curvatures for three representative ribbon shapes

More specifically, this subsection considers planar beams with three representative shapes (sinusoidal, polynomial and arc shapes). Their resulting 3D geometries have important engineering applications, e.g., for uses in tunable electromagnetic devices and stretchable electrical interconnects (Jang et al., 2017; Xu et al., 2015). The corresponding shape functions are given by

$$F_{\text{sinusoidal}}(S, b) = \sin \frac{2\pi S}{L_S}, \quad F_{\text{polynomial}}(S, b) = 64 \left(\frac{1}{2} + \frac{S}{L_S} \right)^3 \left(\frac{1}{2} - \frac{S}{L_S} \right)^3, \quad F_{\text{arc}}(S, b) = \frac{1}{b^2} \left(\cos \frac{bS}{L_S} - \cos \frac{b}{2} \right). \quad (66)$$

According to the aforementioned discussions on Eq. (22), the maximum values of the parameter b are obtained as: $b_{\max} = 0.159$ for $F_{\text{sinusoidal}}(S, b)$, $b_{\max} = 0.291$ for $F_{\text{polynomial}}(S, b)$, $b_{\max} = \pi$ for $F_{\text{arc}}(S, b)$. Substitution of these shape functions into Eqs. (52)–(55) gives the primary terms of displacements and curvatures during postbuckling. Noting that the expressions of b^0 order terms are already given in Eqs. (52)–(55), only the nonzero expressions of b^1 order terms are provided here, i.e.,

$$\begin{aligned} \phi_{(1)b(1)} &= \int \frac{2\pi}{L_S} \left[-\pi^2 + 3(1+\nu) \left(-1 + \cos \frac{2\pi S}{L_S} + \frac{2\pi S}{L_S} \sin \frac{2\pi S}{L_S} \right) \right] dS, \\ U_{2(2)b(1)} &= \frac{L_S}{144} \left\{ \frac{18\pi S}{L_S} \left[-21 - 2\pi^2 + 8\pi^2 \left(\frac{S}{L_S} \right)^2 \right] (1+\nu) + \frac{72\pi S (\pi^2 - 6 - 6\nu)}{L_S} \cos \frac{2\pi S}{L_S} \right. \\ &\quad \left. - \frac{54\pi S (1+\nu)}{L_S} \cos \frac{4\pi S}{L_S} + 9(\pi^2 + 48 + 48\nu) \sin \frac{2\pi S}{L_S} + 81(1+\nu) \sin \frac{4\pi S}{L_S} - 7\pi^2 \sin \frac{6\pi S}{L_S} \right\}, \\ \hat{\kappa}_{3(1)b(1)} &= \frac{2\pi}{L_S} \left[-\pi^2 + 3(1+\nu) \left(-1 + \cos \frac{2\pi S}{L_S} + \frac{2\pi S}{L_S} \sin \frac{2\pi S}{L_S} \right) \right], \end{aligned}$$

for sinusoidal-shaped ribbons, and

$$\phi_{(1)b(1)} = \int \frac{\pi^2 (1+\nu)}{12L_S} \left\{ \frac{12S}{L_S} \cos \frac{2\pi S}{L_S} + \pi \left[-1 + 12 \left(\frac{S}{L_S} \right)^2 \right] \sin \frac{2\pi S}{L_S} \right\} dS,$$

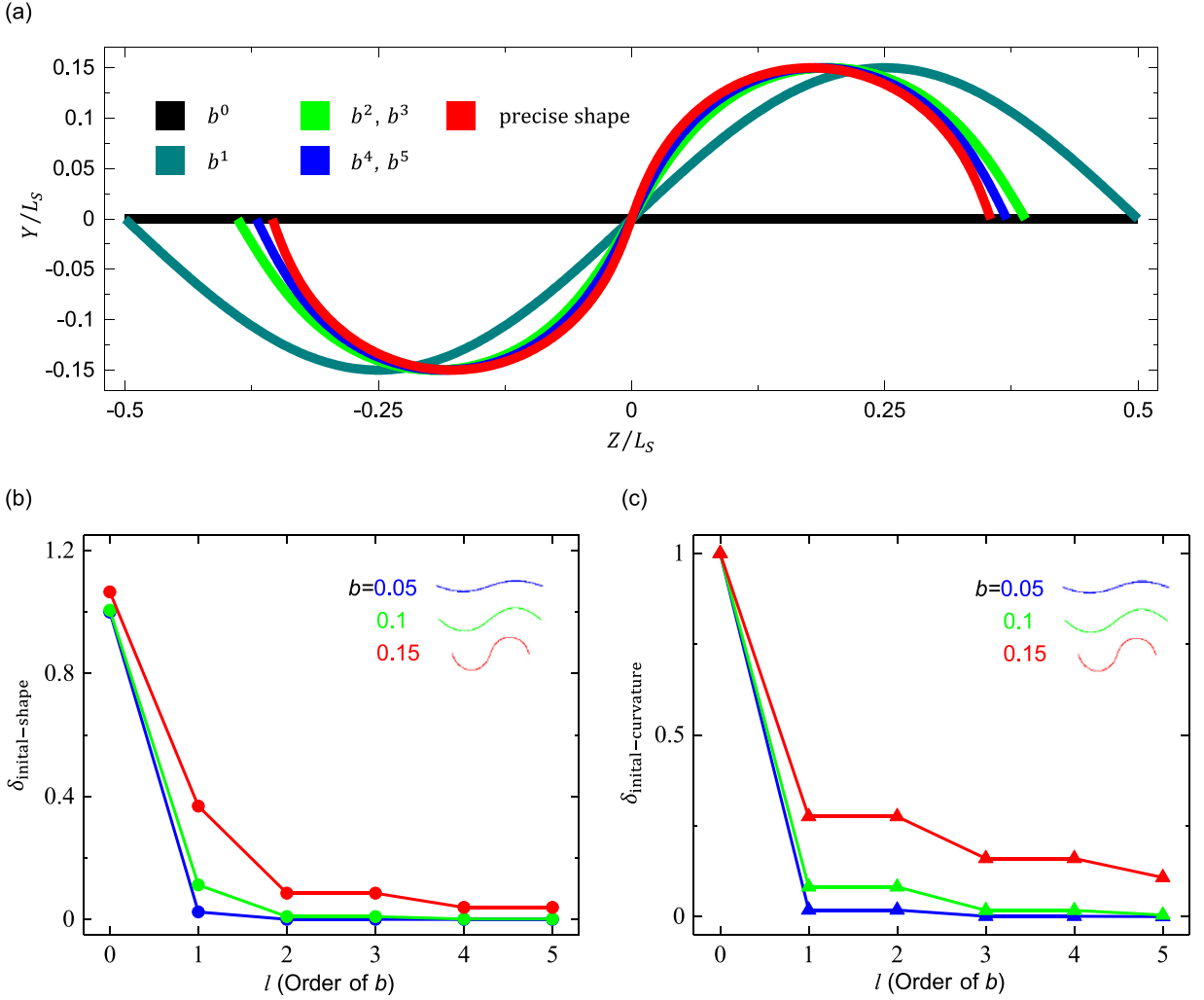


Fig. 5. Relative errors of the initial configuration induced by the double perturbation method: (a) the precise initial configuration of the sinusoidal-shaped beam with $b = 0.15$, and the approximate ones used in the perturbation method with different orders of expansion with respect to the parameter b . Here, the results for expansions b^2 and b^3 are the same, and those for expansions b^4 and b^5 are the same. (b) and (c) the relative errors of initial shape and initial curvature versus the order with respect to the parameter b for the sinusoidal-shaped beam with $b = 0.05, 0.1$ and 0.15 .

$$U_{2(2)b(1)} = \frac{L_S}{768} \left\{ \begin{aligned} & -9 + 135\nu + 2\pi^4(1+\nu) \left(1 - \frac{4S^2}{L_S^2}\right)^2 - 4\pi^2 \left[7 - 5\nu + \frac{48S^2(1+\nu)}{L_S^2}\right] \\ & + \frac{72\pi(1+\nu)S}{L_S} \sin \frac{4\pi S}{L_S} - 32[-6\nu + \pi^2(1+\nu)] \cos \frac{2\pi S}{L_S} \\ & + \left[9 + 57\nu - 2\pi^2(1+\nu) \left(-1 + \frac{12S^2}{L_S^2}\right)\right] \cos \frac{4\pi S}{L_S} \end{aligned} \right\},$$

$$\hat{\kappa}_{3(1)b(1)} = \frac{\pi^2(1+\nu)}{12L_S} \left\{ \frac{12S}{L_S} \cos \frac{2\pi S}{L_S} + \pi \left[-1 + 12 \left(\frac{S}{L_S}\right)^2 \right] \sin \frac{2\pi S}{L_S} \right\},$$

for arc-shaped ribbons. The results for polynomial-shaped ribbons are given in Supporting Information.

From these explicit expressions, it is clear that the small size ratios w/L_S and h/w (e.g., $\ll 0.1$) play negligible roles on the postbuckling deformations. The deformed configuration is determined mainly by the shape function $F(S, b)$, the value of the parameter b , the applied strain ε_{app} , and the Poisson ratio ν . Thus, the scaling law of the maximum strain ε_{max} in Eq. (64) can be rewritten as

$$\varepsilon_{max} \sim \frac{h}{L_S} \sqrt{\varepsilon_{app}} b^0. \quad (67)$$

It dictates the proportional dependence of the maximum strain ε_{max} on the dimensionless height h/L_S .

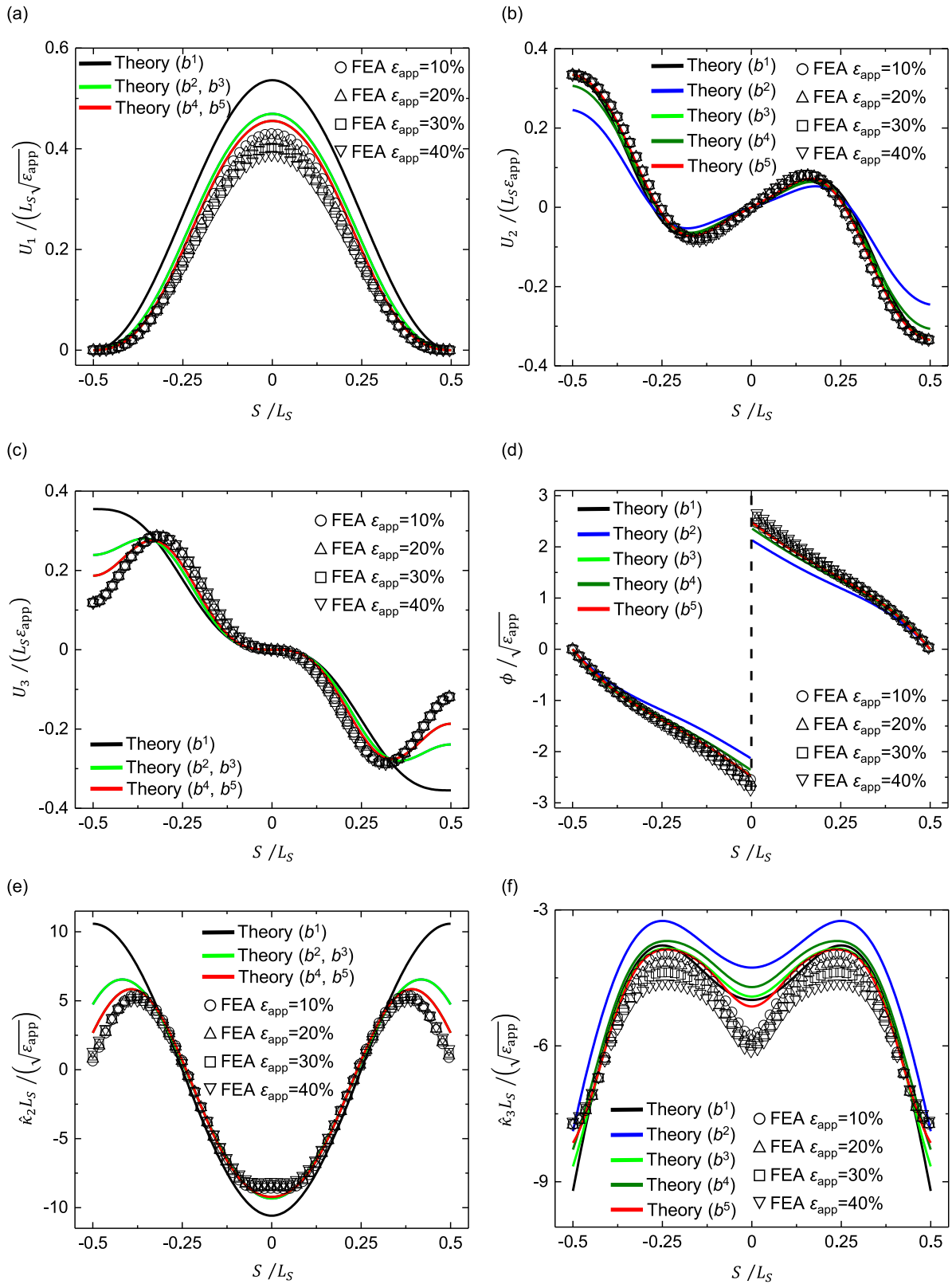


Fig. 6. Theoretical and FEA results on the distributions of the normalized (a)–(c) displacement components, (d) twist angle, (e) bending curvature, and (f) twisting curvature, for the sinusoidal-shaped beam ($b = 0.15$) under the applied strain (10%, 20%, 30%, and 40%). The theoretical predictions contain the results up to b^5 order.

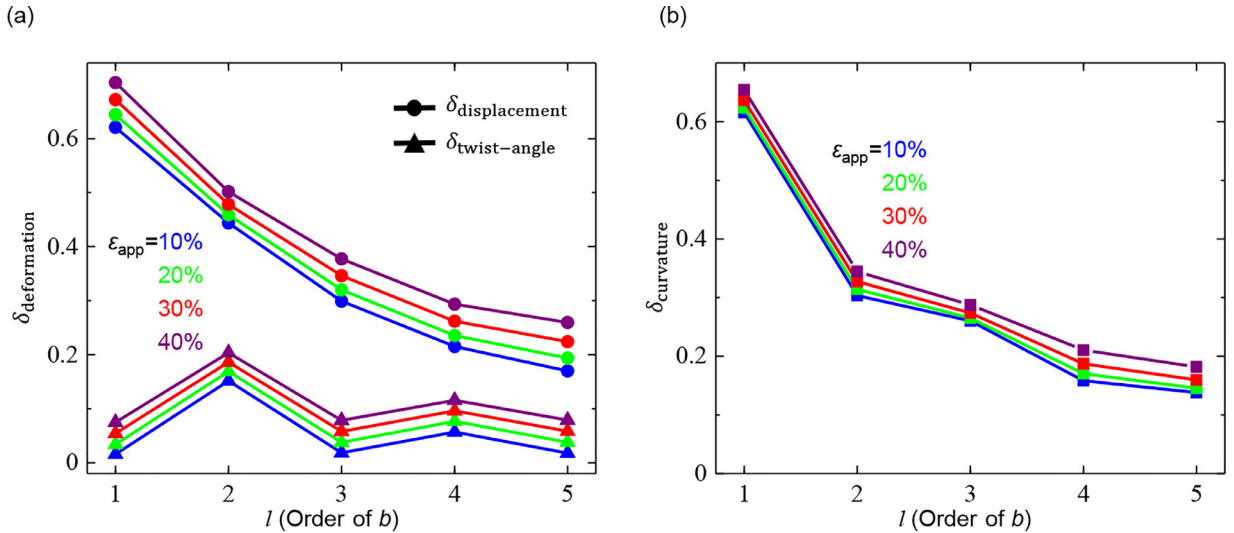


Fig. 7. Relative errors of the predicted deformation and curvature during postbuckling as compared to the FEA results of a sinusoidal-shaped beam ($b = 0.15$) under $\epsilon_{\text{app}} = 10\%$, 20% , 30% , and 40% . (a) Relative errors of the predicted displacement and twist angle versus the order with respect to the parameter b ; (b) Similar results for the predicted curvature.

3.3. Validation of analytic solutions

Both the experiments and FEA (See Supporting Information for more details) were carried out to validate the analytic models described above. A thin film ($40\mu\text{m}$) in polyethylene terephthalate (PET; $E = 3.5\text{GPa}$, $\nu = 0.39$) was patterned into different representative shapes described above, and then integrated with a prestretched silicone elastomer to achieve buckling-guided 3D assembly. Under an end-to-end compression ($\epsilon_{\text{app}} = 30\%$), the deformed configurations of three different shaped ribbons, all with $h : w : L_S = 1 : 30 : 900$, are obtained by experiment, theory and FEA, as illustrated in Fig. 3(a). The analytic predictions agree well with both experiments and FEA results for all three cases. Furthermore, the dimensionless displacement data U_i/L_S were collected from the experiments for quantitative comparison. Fig. 3(b)–(d) provide representative results on a sinusoidal-shaped ribbon under three different levels of applied strain ϵ_{app} . Good agreement can be observed for three displacement components, although slight deviations between the theoretical and experiment results can be observed at $\epsilon_{\text{app}} = 30\%$. This mainly originates from the truncation error of the perturbation method that increases with increasing the parameter a , and thereby the applied strain. Note that the analytic results for the above examples keep almost unchanged when the number of expansion terms with regard to the parameter b exceeds 2.

Under an oblique compression ($\epsilon_{\text{app}} = 30\%$) with two different loading angles ($\alpha = 10^\circ$ and 30°), the sinusoidal-shaped ribbons ($h : w : L_S = 1 : 30 : 900$) of $b = 0.1$ were investigated in details. As shown by Fig. 4(a), the analytic predictions of the deformed configurations also agree well with the results of experiment and FEA. Again, the predicted displacement components U_i/L_S show quantitative agreements with the experiment.

To illustrate the effect of substrate deformations on the assembled ribbon configurations, a full FEA model was established using commercial software ABAQUS, in which solid elements (C3D8R) and shell elements (S4R) with refined meshes were adopted for the 2D precursor and the elastomer substrate, respectively. The results of a representative sinusoidal-shaped ribbon ($b = 0.15$) are presented in Fig. S1 (Supporting Information), in which both the models with and without substrate are taken into account. Here, a relatively large applied strain (40%) is considered, approximately equivalent to the compression induced by use of $\sim 66.7\%$ prestrain in an elastomer substrate. Although slight bending deformations occur in the bonding regions when the substrate deformations are modeled, the resulting 3D deformations of the ribbon-shaped structure keep almost unchanged as compared to the FEA model without any substrate. This is also in accordance with our previous publications (Liu et al., 2016; Xu et al., 2015), in which the predictions of FEA without consideration of substrate deformations agreed well with the experimental measurements.

3.4. Error analysis

After validating the analytic models, this subsection discusses systematically the effect of expansion order with regard to the parameter b on the precision of analytic predictions.

The error associated with the perturbation method results from the truncation of high order terms. In particular, the effect of the higher order terms with respect to the parameter a was discussed for the postbuckling of straight beams (Su et al., 2012). Typically, the use of two expansion terms with respect to the parameter a can give results of acceptable accuracy. Here, we focus on the error resulted from the neglect of higher order terms with respect to the parameter b .

The truncation error of the initial configuration is illustrated in Fig. 5 for a sinusoidal-shaped ribbon. As illustrated in Fig. 5(a), the asymptotic shapes approach the precise shape rapidly with increasing the number of perturbation terms. To quantify the error in describing the initial configuration, two relative errors were introduced, i.e.,

$$\delta_{\text{initial-shape}} = \sqrt{\frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l b^k Y_{b(k)}(S) - Y(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [Y(S)]^2 dS} + \frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l b^k Z_{b(k)}(S) - Z(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [Z(S)]^2 dS}}, \quad (68)$$

$$\delta_{\text{initial-curvature}} = \sqrt{\frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l b^k K_{1b(k)}(S) - K_1(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [K_1(S)]^2 dS}}.$$

As shown in the Fig. 5(b) and (c), these relative errors both reduce with the increase in the order l and the decrease in the parameter b . For $b = 0.1$ and $l = 3$, these relative errors are very close to zero ($\delta_{\text{initial-shape}} \sim 0.906\%$, $\delta_{\text{initial-curvature}} \sim 1.73\%$), indicating that $l = 3$ is sufficient for characterizing the initial configuration with $b < 0.1$. For relative large b (e.g., 0.15), a high perturbation term (e.g., $l = 5$) could be useful for enhancing the precision.

Then the truncation error of the deformed configurations is analyzed by taking the sinusoidal-shaped ribbon under an end-to-end compression as an example. According to the scaling law of the key components in Eqs. (62) and (63), the distributions of normalized displacement and curvature components are presented in Fig. 6. Here, the twist angle $\phi = \int \hat{\kappa}_3 dS$ is calculated using a piecewise function,

$$\phi = \begin{cases} \int_{-\frac{L_S}{2}}^S \hat{\kappa}_3(\xi) d\xi & S \in [-\frac{L_S}{2}, 0) \\ \int_{\frac{L_S}{2}}^S \hat{\kappa}_3(\xi) d\xi & S \in (0, \frac{L_S}{2}] \end{cases}. \quad (69)$$

The theoretical results approach the FEA results gradually by adding more terms with respect to the parameter b in the perturbation solutions. The FEA results for different levels of applied strain also provide further evidence on the scaling law in terms of the applied strain ε_{app} .

To measure the errors of these deformation and curvature components, the following relative errors were defined,

$$\delta_{\text{displacement}} = \sqrt{\frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l ab^k U_{1(1)b(k)}(S) - U_1^{\text{FEA}}(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [U_1^{\text{FEA}}(S)]^2 dS} + \frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l a^2 b^k U_{2(2)b(k)}(S) - U_2^{\text{FEA}}(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [U_2^{\text{FEA}}(S)]^2 dS} + \frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l a^2 b^k U_{3(2)b(k)}(S) - U_3^{\text{FEA}}(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [U_3^{\text{FEA}}(S)]^2 dS}}, \quad (70)$$

$$\delta_{\text{twist-angle}} = \sqrt{\frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l ab^k \phi_{(1)b(k)}(S) - \phi^{\text{FEA}}(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\phi^{\text{FEA}}(S)]^2 dS}},$$

$$\delta_{\text{curvature}} = \sqrt{\frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l ab^k \hat{\kappa}_{2(1)b(k)}(S) - \hat{\kappa}_2^{\text{FEA}}(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\hat{\kappa}_2^{\text{FEA}}(S)]^2 dS} + \frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\sum_{k=0}^l ab^k \hat{\kappa}_{3(1)b(k)}(S) - \hat{\kappa}_3^{\text{FEA}}(S)]^2 dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} [\hat{\kappa}_3^{\text{FEA}}(S)]^2 dS}}.$$

These relative errors were calculated for the sinusoidal-shaped ribbon under $\varepsilon_{\text{app}} = 10\%$, 20%, 30%, and 40%, as shown in Fig. 7. Note that the prestrain of elastomer substrate used in most of the previous experiments on the buckling-guided 3D assembly (see for example, Xu et al., 2015, Yan et al., 2016a) is typically below 70%, corresponding to a compressive of $\sim 41.2\%$. The relative errors ($\delta_{\text{displacement}}$ and $\delta_{\text{curvature}}$) of displacements and curvatures decrease monotonically with increasing the number of perturbation terms or decreasing the applied strain ε_{app} . Differently, the results for the twist angle ($\delta_{\text{twist-angle}}$) have a minor fluctuation that stays at a relative low level ($< 11.6\%$) for $l > 2$, which can be attributed to the odeivity of the twist angle ϕ in the 4th equation of Eq. (58).

3.5. Mode ratio and maximum strain

According to Xu et al. (2015), the mode ratio R_{mode} serves as an important parameter to characterize the deformation mode of different 3D ribbon mesostructures from buckling-guided assembly. Substitution of Eq. (59) into Eq. (65) gives the

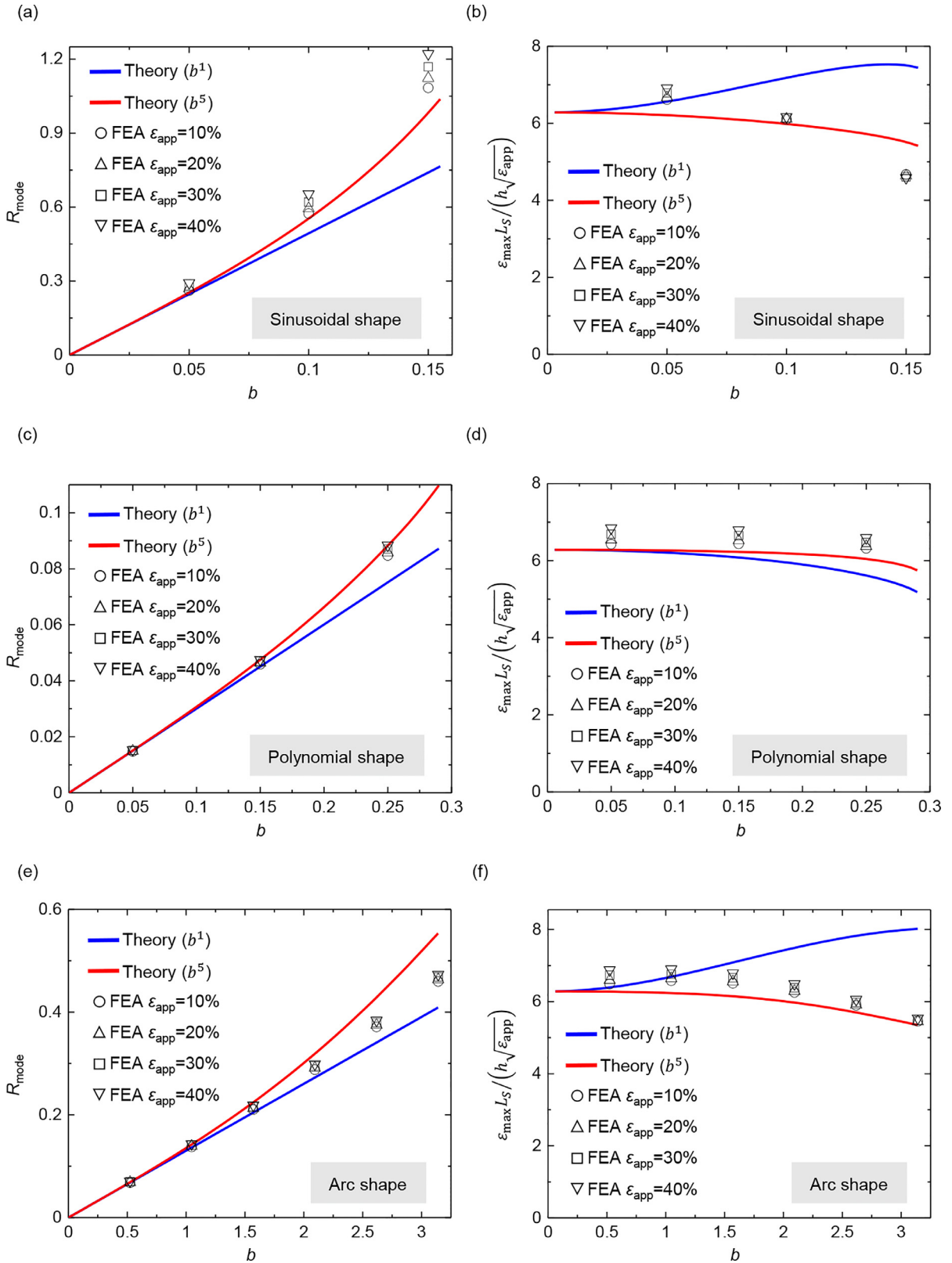


Fig. 8. Theoretical and FEA results of the mode ratio R_{mode} and the normalized maximum strain $\epsilon_{max} L_S / (h \sqrt{\epsilon_{app}})$ versus the value of the parameter b for the planar curved beams ((a) & (b) sinusoidal shape; (c) & (d) polynomial shape; and (e) & (f) arc shape) under the applied strain $\epsilon_{app} = 10\%$, 20% , 30% , and 40% .

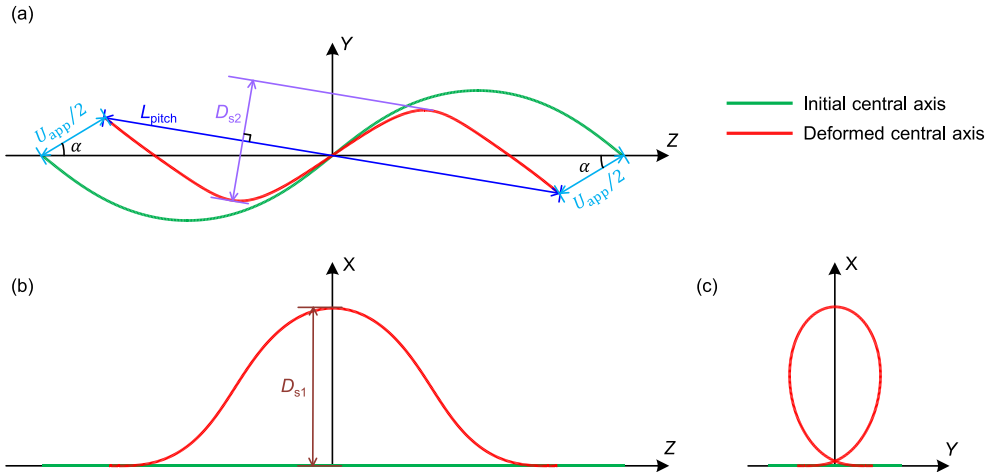


Fig. 9. Schematic illustration of the initial and deformed central axes of a sinusoidal-shaped beam under a general compression.

expansion form of the mode ratio R_{mode} as

$$R_{mode} \approx \frac{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} \left| \sum_{k=0}^l b^k \hat{k}_{3(1)b(k)}(S) \right| dS}{\int_{-\frac{L_S}{2}}^{\frac{L_S}{2}} \left| \sum_{k=0}^l b^k \hat{k}_{2(1)b(k)}(S) \right| dS}. \quad (71)$$

For the three representative ribbons shapes discussed in the above sections, the predicted mode ratios and FEA results are illustrated in Fig. 8(a), (c), and (e), for the parameter b in the range of $[0, b_{max}]$. When b is relatively small (e.g., $b \leq b_{max}/3$), the mode ratio shows a proportional dependence on the parameter b , with the coefficient of proportionality given by $\pi^2/2$ for sinusoidal shape, $3(5040\pi^2 - 33600 - 140\pi^4 - \pi^6)(1 + \nu)/(70\pi^5)$ (or $0.216(1 + \nu)$) for polynomial shape, and $(24 - \pi^2)(1 + \nu)/(48\pi)$ (or $0.0937(1 + \nu)$) for arc shape. When b is relative large, these proportional relations underestimate the magnitude of the mode ratio. The peak value of mode ratio accessible to the different ribbon-shaped designs can be used to identify the dominated type of deformations during the postbuckling process. For example, the peak mode ratio of the polynomial-shaped beam is only ~ 0.1 , suggesting that the assembled 3D mesostructure is always bending-dominated. In contrast, the mode ratio exceeds 0.2 for $b > 0.0398$ and approaches ~ 1.08 at b_{max} for the sinusoidal-shaped beam, indicating the assembled 3D mesostructure experiences a mode of mixed bending and twisting deformations in this condition.

The peak value ε_{max} of maximum principal strain in the ribbon is also an important quantity in the design of mechanically-guided 3D assembly, since an excessive ε_{max} can induce structural failure. Using Eq. (60), the normalized maximum strain can be given as $\varepsilon_{max} L_S / (h \sqrt{\varepsilon_{app}})$. The analytic predictions of this normalized maximum strain and the FEA results are shown in Fig. 8(b), (d), (f). Despite a slight deviation for the sinusoidal-shaped ribbon at $b = 0.15$, the theoretical results up to 5th order of the parameter b agree reasonably well with the FEA results. In addition, the normalized maximum strain is found to be change very slightly during the entire range of $[0, b_{max}]$ for the polynomial- and arc-shaped designs, which is, thereby, approximately the same as that (2π) of a straight ribbon. This is mainly due to the bending-dominated deformation for most of these geometries. It is also noteworthy that most of the predicted deformation quantities (i.e., displacement components, twist angle, curvature components, mode ratio and maximum principal strains) in Figs. 6–8 are within 20% difference (in relative) as compared to the FEA results, even for the applied strain up to 40%.

3.6. Inverse design of 3D helical mesostructures

The double perturbation method introduced above can establish the relationship between the final 3D geometry and initial 2D ribbon shape, thereby offering a theoretical reference of the inverse design. This section presents an example in the inverse design of 3D helical mesostructures that have widespread applications in electronics (e.g., inductors, antennas), bio-integrated devices (Huang et al., 2012; Jang et al., 2017; Karnaushenko et al., 2015), MEMS (Dean et al., 2000; Farahani et al., 2012), etc. In these applications, the mechanical and electromagnetic properties are highly dependent on the 3D geometries of the helical mesostructures.

As shown by Fig. 3 and the results of Xu et al. (2015), the lateral buckling of sinusoidal-shaped ribbons leads to the formation of 3D helical mesostructures. Here, the inverse design aims to determine the specific sinusoidal shape of the 2D ribbon and the prestretch associated with the buckling-guided assembly, from which the desired 3D helical mesostructure can be formed. To describe the overall shape of these 3D helical mesostructures, three parameters (i.e., the pitch L_{pitch} , two sectional dimensions D_{s1} and D_{s2}) are introduced, as illustrated in Fig. 9. Under a general compression with a loading angle

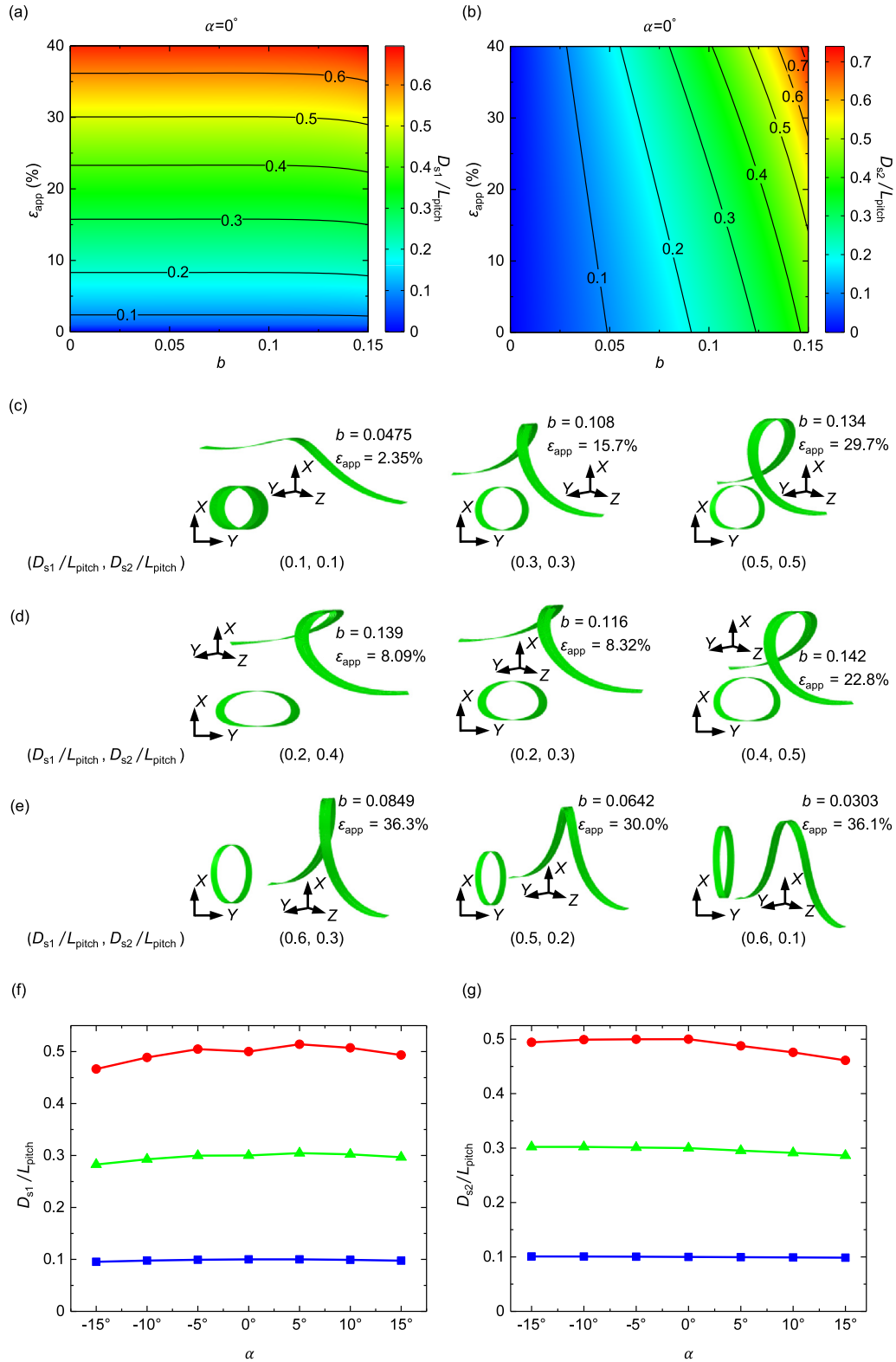


Fig. 10. (a) and (b) Contour plots for the dimensionless shape parameters (D_{s1}/L_{pitch} and D_{s2}/L_{pitch}) of the helical microstructures in terms of the parameter b and the applied strain ϵ_{app} of the end-to-end compression. (c)–(e) Illustration of configurations of the helical microstructures with a variety of shapes: (c) $(D_{s1}/L_{pitch}, D_{s2}/L_{pitch}) = (0.1, 0.1)$, $(0.3, 0.3)$ and $(0.5, 0.5)$; (d) $(D_{s1}/L_{pitch}, D_{s2}/L_{pitch}) = (0.2, 0.4)$, $(0.2, 0.3)$ and $(0.4, 0.5)$; (e) $(D_{s1}/L_{pitch}, D_{s2}/L_{pitch}) = (0.6, 0.3)$, $(0.5, 0.2)$ and $(0.6, 0.1)$. (f) and (g) Dimensionless shape parameters D_{s1}/L_{pitch} and D_{s2}/L_{pitch} versus the loading angle α . In (f) and (g), the square, triangle and circle symbols correspond to the results based on $(b = 0.0475$ and $\epsilon_{app} = 2.35\%)$, $(b = 0.108$ and $\epsilon_{app} = 15.7\%)$ and $(b = 0.134$ and $\epsilon_{app} = 29.7\%)$, respectively.

(α), these shape parameters can be correlated to the displacement components (U_1, U_2 and U_3) by

$$\begin{aligned} L_{\text{pitch}} &= L_Z \sqrt{1 - 2\varepsilon_{\text{app}} \cos^2 \alpha + \varepsilon_{\text{app}}^2 \cos^2 \alpha}, \\ D_{s1} &= U_1|_{S=0}, \\ D_{s2} &= 2 \max \left[\begin{aligned} &\frac{1 - \varepsilon_{\text{app}} \cos^2 \alpha}{\sqrt{(1 - \varepsilon_{\text{app}} \cos^2 \alpha)^2 + (\varepsilon_{\text{app}} \cos \alpha \sin \alpha)^2}} [Y(S) + U_2(S) \cos \Theta + U_3(S) \sin \Theta] \\ &+ \frac{\varepsilon_{\text{app}} \cos \alpha \sin \alpha}{\sqrt{(1 - \varepsilon_{\text{app}} \cos^2 \alpha)^2 + (\varepsilon_{\text{app}} \cos \alpha \sin \alpha)^2}} [Z(S) - U_2(S) \sin \Theta + U_3(S) \cos \Theta] \end{aligned} \right]. \end{aligned} \quad (72)$$

For a prescribed 3D helical mesostructure characterized by two dimensionless shape parameters (D_{s1}/L_{pitch} and D_{s2}/L_{pitch}), the parameter b , the applied strain ε_{app} and the loading angle (α) can be then tailored precisely to yield the desired geometry. In principle, the solution to such inverse problem is not unique, since only two main shape parameters are prescribed here.

For the most representative condition of buckling-guided assembly, the ribbons undergo an end-to-end compression ($\alpha = 0$). In this condition, the effects of the parameter b and the applied strain ε_{app} on the dimensionless ratios D_{s1}/L_{pitch} and D_{s2}/L_{pitch} are illustrated in Fig. 10(a) and (b). These results suggest that the shape parameter D_{s1}/L_{pitch} is almost independent on the parameter b , and thereby, can be approximated by the counterpart of a straight ribbon, i.e.,

$$\left. \frac{D_{s1}}{L_{\text{pitch}}} \right|_{b=0} = \frac{2\sqrt{\varepsilon_{\text{app}}}}{\pi(1 - \varepsilon_{\text{app}})}. \quad (73)$$

The shape parameter D_{s2}/L_{pitch} is not quite sensitive to the change in the applied strain. For the parameters ($\varepsilon_{\text{app}} \leq 40\%$ and $b \leq 0.15$) of interest herein, D_{s1}/L_{pitch} and D_{s2}/L_{pitch} can reach 0.694 and 0.737, respectively. Given a desired shape of the helical mesostructure, the required 2D ribbon shape and the applied strain can be then easily determined. For example, a set of helical mesostructures with almost circular sections can be obtained from the above inverse design model, with the shape parameters $D_{s1}/L_{\text{pitch}} = D_{s2}/L_{\text{pitch}} = 0.1, 0.3$, and 0.5 , as shown in Fig. 10(c). A different set of helical mesostructures with elliptical sections of diverse length ratios between semimajor and semiminor axes can be also achieved, as shown in Figs. 10(d) and (e).

During the assembly experiments, it is typically challenging to maintain strict end-to-end compression, especially under uniaxial stretching, due to the imperfect alignment. In this condition, a slight level of oblique compression can be induced, e.g., with the loading angle from -15° to 15° . For those examples shown in Fig. 10(f) and (g), such level of deviation in the loading angle plays a minor role on the final 3D shapes, with the maximum variation in D_{s1}/L_{pitch} and D_{s2}/L_{pitch} within 7.79%. This suggests that the slight level of oblique compression that might be encountered in experiments has little influence on the 3D geometry of buckling assembly.

4. Conclusions

In summary, this paper presents a double perturbation method for the postbuckling analyses of planar curved beams of relevance to techniques in mechanically-guided 3D assembly. By introducing two perturbation parameters related to the initial configuration and the out-of-plane displacement, the highly nonlinear governing equations can be transformed into a series of linear differential equations. Analytical solutions of the deformations, curvatures and maximum strain during postbuckling are obtained for relatively general ribbon shapes. Experiments and FEA on three representative shapes, including sinusoidal, polynomial and arc configurations, illustrate the validity of the developed method. The results on the effect of the expansion order with respect to the perturbation parameter b elucidate the convergence of the method. The theoretical results based on the expansion up to a^2b^1 provide acceptable accuracy for relatively small parameter b (i.e., $< b_{\text{max}}/3$). The normalized maximum strains ($\varepsilon_{\text{max}} L_S / (h\sqrt{\varepsilon_{\text{app}}})$) of polynomial- and arc-shaped beams were found to be insensitive to the magnitude of b . Finally, an inverse design problem for a class of widely used helical geometries illustrates the utility of the double perturbation method. It was found that the shape parameter D_{s1}/L_{pitch} related to the out-of-plane dimension can be approximated by the counterpart of a straight ribbon. This method can be potentially extended to the inverse design of general 3D ribbon shapes. These results and the general method can serve as important design references for the development of 3D functional micro-devices using the buckling-guided assembly.

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Appendix A. Higher-order expressions of curvatures and cosine coefficients

The expressions of $\hat{\kappa}_i^{(2)}$ and $a_{ij}^{(2)}$:

$$\left\{ \begin{aligned} \hat{\kappa}_1^{(2)} &= \left\{ \begin{aligned} &\psi U''_1 + (U'_3 U'_2)' + (U_2 U'_2 - U_3 U'_3) K'_1 \\ &+ \left[(U_2 U''_2 - U_3 U''_3) + (U'^2_2 - U'^2_3) \right] K_1 - (U_2 U_3)' K^2_1 \end{aligned} \right\}, \\ \hat{\kappa}_2^{(2)} &= \left[\begin{aligned} &\psi U''_2 - (U'_3 U'_1)' - (\psi U_3 + U'_1 U_2) K'_1 \\ &- (\psi U'_3 + \varphi + U''_1 U_2 + \frac{3}{2} U'_1 U'_2) K_1 + \frac{1}{2} U'_1 U_3 K^2_1 \end{aligned} \right] \\ \hat{\kappa}_3^{(2)} &= 0 \end{aligned} \right. \quad (A.1)$$

$$\left\{ \begin{aligned} a_{11}^{(2)} &= -\frac{1}{2} (\psi^2 + U'^2_1) \\ a_{12}^{(2)} &= -\frac{1}{2} U'_1 U'_2 + \frac{1}{2} U'_1 U_3 K_1 + \varphi \\ a_{13}^{(2)} &= U'_3 U'_1 - \psi U'_2 + (U'_1 U_2 + \psi U_3) K_1 \end{aligned} \right. \quad (A.2)$$

$$\left\{ \begin{aligned} a_{21}^{(2)} &= -\frac{1}{2} U'_1 U'_2 + \frac{1}{2} U'_1 U_3 K_1 - \varphi \\ a_{22}^{(2)} &= -\frac{1}{2} \left[\psi^2 + (U'_2 - U_3 K_1)^2 \right] \\ a_{23}^{(2)} &= \psi U'_1 + (U'_3 + U_2 K_1) (U'_2 - U_3 K_1) \end{aligned} \right. \quad (A.3)$$

$$\left\{ \begin{aligned} a_{31}^{(2)} &= -(U'_3 + U_2 K_1) U'_1 \\ a_{32}^{(2)} &= -(U'_3 + U_2 K_1) (U'_2 - U_3 K_1) \\ a_{33}^{(2)} &= -\frac{1}{2} \left[U'^2_1 + (U'_2 - U_3 K_1)^2 \right] \end{aligned} \right. \quad (A.4)$$

in which $(\cdot)' = d(\cdot)/dS$, and

$$\varphi = \int \left[\frac{1}{2} (U'_1 U''_2 - U''_1 U'_2) - U'_1 U_3 K'_1 + \frac{1}{2} (U'_1 U_3 K_1)' + U'_1 U_2 K^2_1 \right] dS.$$

Supporting Information

Supporting Information associated with this article can be found, in the online version, at [doi:10.1016/j.jmps.2017.10.012](https://doi.org/10.1016/j.jmps.2017.10.012).

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